

IndiGo Enterprise Digital Analytics Platform

A unified analytics ecosystem — Contentsquare, GA4, and WebEngage — our response to the IndiGo Enterprise Digital Analytics Platform RFP.

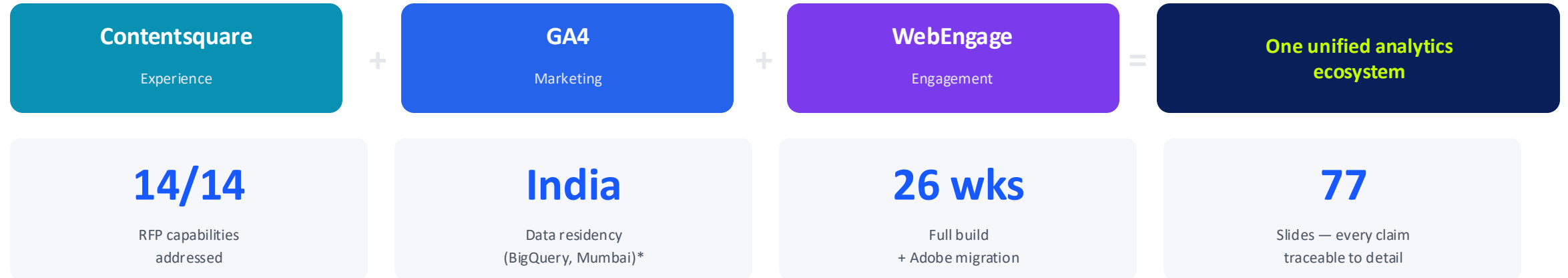
Prepared for the IndiGo evaluation committee

Axeno Consulting · Response Version 2.0 · July 2026

Agenda

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|-----------|---|-----------|--|
| 01 | Understanding of context & objectives | 06 | Implementation approach & timeline |
| 02 | Solution vision & architecture | 07 | Delivery confidence — Implementation, cutover, team & risk |
| 03 | Functional scope coverage & user journeys | 08 | Team, support track & commercial structure |
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| 05 | Extended capability roadmap | | |

The proposal in one glance



WHY THREE PLATFORMS, NOT ONE: No single vendor natively covers all the requirements mentioned in the RFP (Product analytics + experience analytics + marketing attribution + engagement/CDP). Our recommendation delivers category-leading depth on all 14 capabilities instead of shallow coverage forced by a single-vendor mandate.

FOR THE BUSINESS

- ✓ Every booking, check-in, disruption & loyalty journey instrumented end-to-end
- ✓ One data layer - no more reconciling numbers across tools
- ✓ 26-week delivery with 5 IndiGo sign-off gates, not a black box
- ✓ Fixed-price build, transparent pricing model (pricing will be shared once sizing information has been provided)

FOR THE TECH TEAM

- ✓ HLA + LLA architecture, middleware, and data-persistence layers fully specified
- ✓ Every integration classified real-time vs. batch, mapped to actual use cases
- ✓ Terraform IaC, CI/CD, and a named environment-promotion path (Dev→Prod)
- ✓ Full technical detail lives in the Appendix — nothing hidden, just organised

This slide is the summary. Sections 01–09 that follow are the evidence.

*All data from CSQ, GA4 and webengage would be pushed into Bigquery with an India data center. CSQ's data center is in Europe. However, CSQ does not collect any PII data and therefore is compliant to the DPDP rules.

01

Understanding of context & objectives

IndiGo's digital analytics landscape, and what success must look like.

IndiGo's digital analytics opportunity — and our response

The opportunity

IndiGo is India's largest airline - 100M+ annual passengers, 441 aircraft, 142 destinations. As IndiGo transitions to a modern, unified digital analytics ecosystem, it needs a strategic technology partner capable of delivering enterprise-grade analytics at scale.

Axeno Consulting proposes a best-of-breed platform combining three specialised, category-leading technologies — Contentsquare for experience & product analytics, GA4 for marketing attribution, and WebEngage for customer engagement & audience activation.

Our solution addresses all 14 functional capabilities and 9 non-functional requirement areas, with a clear roadmap for extended capabilities and recommended complementary tools.

100M+

Annual passengers across IndiGo's digital channels

14

Functional capability areas addressed in this response

26 wk

End-to-end implementation

3

Best-of-breed platforms unified through Axeno

Three platforms, one accountable partner

WHY A HYBRID, MULTI-VENDOR ARCHITECTURE — NOT A SINGLE PLATFORM

IndiGo's 14 capabilities span three genuinely different disciplines — digital experience analytics, marketing attribution, and customer engagement/CDP. No platform in the market natively excels at all three: Contentsquare has no marketing attribution engine or CDP; GA4 has no session replay, heatmaps, or frustration detection; WebEngage has no digital experience analytics. Even Adobe's closest single-vendor answer requires stitching together 3–4 separate Adobe products at materially higher cost, and still falls short of Contentsquare's depth in behavioural diagnostics. A single-vendor mandate would force IndiGo to accept shallow coverage on 8–9 of its 14 capabilities — this hybrid architecture instead delivers category-leading depth on every one of them, with the resulting integration complexity absorbed by Axeno's middleware (CDP) and unified data layer (BigQuery), not left for IndiGo to manage.

Contentsquare

Experience & product analytics

- Session replay & heatmaps
- Frustration detection (rage/dead clicks)
- Journey analysis (21-step Sunburst)
- Product analytics (autocapture)
- Booking funnel optimisation
- Sense AI — natural language analytics
- Impact Quantification (revenue per element)
- Cross-device attribution
- Unified customer profiles

Google Analytics 4

Marketing attribution

- Multi-touch attribution (6 models)
- Campaign performance (ROAS, CAC)
- Customer Lifetime Value modelling
- Acquisition channel analysis
- BigQuery native export
- Predictive metrics (churn, purchase probability)

WebEngage

Engagement & audience activation

- Dynamic audience segmentation
- Journey orchestration & automation
- Push / Email / SMS / WhatsApp / In-app
- Predictive segmentation (ML-based)
- Campaign analytics & CLV tracking

Why this combination wins for IndiGo

01

Airline-proven DXA

Contentsquare is used by leading airlines like Ryanair and easyJet. It uniquely combines product analytics and experience analytics to give specific and actionable behavioral intelligence for CRO and experience improvement

02

Industry-standard attribution

GA4 provides comprehensive attribution models, ROAS/CAC measurement, and native BigQuery integration — the marketing analytics depth that completes the stack. Data-driven attribution across all channels.

03

Audience activation, unified

WebEngage extends the stack with CDP-grade identity resolution and full-channel engagement — push, email, SMS, WhatsApp, in-app — with India hosting, orchestrating audiences that Contentsquare's rich behavioural signals help define.

04

Execution by Axeno

Delhi-based team with deep technology experience of bringing together solutions and Adobe migration expertise. On-ground delivery and complete accountability of performance. Single escalation path through Axeno.

05

India data residency via BigQuery

BigQuery on GCP Mumbai (asia-south1) as India-resident data warehouse, unifying all three platforms for enterprise BI and custom ML models via Vertex AI. CSQ's data center is in Europe. However, CSQ does not collect any PII data and therefore is compliant to the DPDP rules.

06

Transparent, extensible design

We map every RFP requirement to the right platform combination and design clear extension paths — Adobe Target, OneTrust, CDP, DataHub — for capabilities that benefit from additional specialised tooling.

02

Solution vision & architecture

14 functional capabilities mapped to platform ownership, complementary tech stack, and RFP objectives alignment.

14 RFP capabilities mapped to platform ownership

#	Capability area	Primary platform	Secondary	Additional stack
1	Enterprise Data Collection	Contentsquare	Adobe Analytics/GA4 + WebEngage	GTM, CDP
2	Event Taxonomy & Governance	Contentsquare	Adobe Analytics/GA4	GTM, CDP
3	Identity Resolution & Profiles	Contentsquare	WebEngage	CDP
4	Customer Journey Analytics	Contentsquare	WebEngage	BigQuery
5	Product Analytics	Contentsquare	WebEngage / BigQuery	Adobe Analytics/GA4
6	Booking Funnel & CRO	Contentsquare	Adobe Analytics/GA4	Adobe Target/Equivalent
7	Marketing Analytics & Attribution	Adobe Analytics/GA4	Contentsquare	BigQuery, Looker
8	Audience Mgmt & Segmentation	WebEngage	Contentsquare	CDP
9	AI-Powered & Predictive Analytics	Adobe Analytics/GA4	Contentsquare	BigQuery ML
10	Digital Experience Analytics	Contentsquare	—	existing APM/Observability
11	Heatmaps & Interaction Analytics	Contentsquare	—	—
12	Reporting & Dashboards	BigQuery	Contentsquare	Looker / Power BI
13	Self-Service Analytics	Contentsquare	Adobe Analytics/GA4 + WebEngage	Looker/Power BI
14	Data Quality & Governance	Contentsquare	Adobe Analytics/GA4	CDP, DataHub

Additional components extending the stack for full RFP coverage

TAG MANAGEMENT

Free

Google Tag Manager

Event governance, version control, approval workflows, server-side tagging on GCP

DATA WAREHOUSE

GCP

BigQuery (GCP Mumbai)

India-resident unified analytics store for cross-platform data unification and ML

EXPERIMENTATION

Commercial

Adobe Target/Equivalent

A/B testing & conversion rate optimisation. Native CS + GA4 integration. Strong India presence

OBSERVABILITY

Commercial / OSS

Existing APM / Observability platform

Technical monitoring integration — API failures, page performance, error correlation

EVENT PIPELINE

OSS

Existing CDP/Recommended

Schema validation, identity resolution, deduplication, routing to all platforms

ENTERPRISE BI

Free / Licensed

Looker Studio / Power BI

Executive dashboards spanning Contentsquare + GA4 + WebEngage data via BigQuery

CONSENT MGMT

Commercial

OneTrust

DPDP consent management (primary); extends to GDPR for EU-origin passenger data on international routes

DATA CATALOGUE

Open-Source

DataHub

Enterprise data lineage, metadata repository, business glossary management

How each RFP objective maps to our proposed solution

Enterprise Digital Analytics

Contentsquare DXA + CSQ PA - unified behavioural, marketing, and product analytics across web, mobile, kiosks, and backend APIs

Customer Journey Visibility

Contentsquare Journey Analysis (21-step Sunburst) + session replay for diagnostic context + WebEngage cross-channel journey orchestration

Product & Business Analytics

CSQ PA autocapture for feature adoption + BigQuery for unified product health dashboards

Marketing & Campaign Measurement

GA4 multi-touch attribution (6 models) + WebEngage engagement analytics + BigQuery for cross-platform marketing ROI

Data Governance & Compliance

CDP Tracking Plans (schema registry) + GTM governance + OneTrust consent management + BigQuery Mumbai for India data residency

AI & Advanced Analytics

Contentsquare Sense AI (NL analytics, anomaly detection) + GA4 predictive metrics + BigQuery ML / Vertex AI for custom models

Scalability & Future Readiness

All SaaS platforms scale horizontally. BigQuery on GCP provides infinite compute elasticity. Event-driven architecture via CDP

03

Functional scope coverage

Point-by-point response to all 14 RFP capability areas, with platform ownership, detailed approach, and the combined-stack delivery model for each.

Enterprise data collection & event governance

✓ 01 Enterprise Data Collection & Instrumentation			Compliant
APPROACH - Contentsquare Smart Capture auto-collects all interactions across web, app, chatbot conversations, and kiosk sessions — one instrumentation layer, not a separate tool per channel - Adobe Analytics/GA4 tagging for marketing attribution events - WebEngage SDK for engagement events (push, email, SMS, WhatsApp) - Server-side capture for payments, refunds, bookings - Offline queuing with auto-sync across all channels	PLATFORM OWNERSHIP Primary Contentsquare Secondary WebEngage Additional GTM CDP	EXTENDED COVERAGE VIA Chatbot and kiosk analytics are tracked directly through Contentsquare — the same instrumentation used for web/app, not a separate collection mechanism requiring extra integration.	
✓ 02 Event Taxonomy & Data Governance			Compliant — combined stack
APPROACH - CDP Tracking Plans (schema registry) as central schema registry — single source of truth for event names, properties, types - GTM workspace version control and approval workflows for tag deployment - CSQ PA in-platform taxonomy management with naming conventions and metadata - GA4 DebugView for pre-deployment validation - Axeno will design Event Governance Playbook (object, action, naming, ownership, deprecation)	PLATFORM OWNERSHIP Primary Contentsquare Secondary GTM Additional CDP	EXTENDED COVERAGE VIA Maker-checker governance is implemented through GTM workspace approval flows and documented process controls. DataHub extends this with an enterprise metadata catalogue and business glossary.	

Identity resolution & customer journey analytics

✓ 03

Customer Identity Resolution & Unified Profiles

Compliant

APPROACH

- Contentsquare as the primary identity layer — stitches unknown (anonymous) visitors to known identity the moment login/PNR/loyalty context becomes available, building a rich behavioural profile from the first anonymous visit
- WebEngage as secondary — deterministic identity matching (login IDs, BluChip numbers, PNR, email, phone) for persistent, enrichable engagement profiles
- Adobe Analytics/GA4 for cross-device session identity within marketing analytics
- CDP identity resolution for cross-platform canonical user_id
- Profile enrichment from CRM (Salesforce), PSS, loyalty via WebEngage REST API

PLATFORM OWNERSHIP

Primary

Contentsquare

Secondary

WebEngage

Tertiary

Adobe Analytics/GA4

Additional

CDP

EXTENDED COVERAGE VIA

Contentsquare performs the unknown-to-known identity stitch as the primary mechanism; WebEngage extends this into persistent, enrichable profiles for engagement activation. Household/family modelling is delivered via PNR-derived group attributes in WebEngage, with profile change history logged in BigQuery.

✓ 04

Customer Journey Analytics

Compliant – category leader

APPROACH

- Contentsquare Journey Analysis — interactive Sunburst maps up to 21 steps
- Cross-device, cross-session reconstruction with frustration context
- Drop-off analysis with rage click / dead click / error correlation
- WebEngage for cross-channel journey orchestration (email, push, SMS touchpoints)
- Sense AI automatically surfaces highest-impact drop-off points

PLATFORM OWNERSHIP

Primary

Contentsquare

Secondary

WebEngage

Tertiary

GA4

Additional

BigQuery

EXTENDED COVERAGE VIA

IROPS and flight-delay correlation is delivered via a custom BigQuery ETL layer, unifying contact-centre and loyalty-platform touchpoints into the same journey view. All raw behavioural data and its native reporting remain in Contentsquare as the system of record — other platforms consume derived/aggregated signals, not the raw behavioural feed.

Product analytics & booking funnel optimisation

✓ 05

Product Analytics

Compliant – category leader

APPROACH

- Contentsquare Product Analytics: CSQ PA is unique with its autocapture feature— every feature interaction recorded without pre-instrumentation of events
- Feature adoption, retention, cohort analysis, funnel analysis across Android/iOS/Web
- Product usage data pushed downstream to WebEngage (for engagement-triggered actions) or BigQuery (for enterprise reporting), depending on the use case
- Version-over-version engagement comparison after app releases
- Sense AI for natural language analytics questions

PLATFORM OWNERSHIP

Primary

Contentsquare Product Analytics Module

Secondary

WebEngage / BigQuery

Additional

Adobe Analytics/GA4

EXTENDED COVERAGE VIA

Account-based analytics for corporate booking accounts is delivered via custom segmentation using corporate account IDs, layered on top of native CSQ PA capabilities.

✓ 06

Booking Funnel Analytics & Conversion Optimisation

Compliant — category leader

APPROACH

- Contentsquare's strongest airline capability — Search through Confirmation with full behavioural context
- Payment success/failure pattern analysis by method, device, segment
- Ancillary attachment rate monitoring and optimisation
- Real-time funnel monitoring during flash sales and IROPS
- Session replay of abandoned bookings shows exactly what users experienced
- Adobe Target or equivalent for A/B testing funnel changes identified by CS analysis

PLATFORM OWNERSHIP

Primary

Contentsquare

Secondary

GA4

Additional

Adobe Target

EXTENDED COVERAGE VIA

Adobe Target extends this capability with native experimentation, enabling continuous A/B testing of funnel optimisations identified through Contentsquare's analysis. Ryanair, easyJet, and Thalys (499% uplift) are reference customers.

Marketing attribution & audience management



07 Marketing Analytics & Attribution

Compliant – combined stack

APPROACH

- Adobe Analytics/GA4 as the primary attribution engine — 6 attribution models (first-touch, last-touch, linear, time-decay, position-based, data-driven), building on IndiGo's existing Adobe Analytics investment where preferred
- ROAS & CAC measurement via ad-platform cost data import
- CLV analysis using predictive CLV models
- Contentsquare for cross-device attribution specifically — stitching a customer's journey across devices via its own identity layer, independent of ad-platform device graphs
- BigQuery export for enterprise-scale marketing analytics
- WebEngage for engagement channel attribution (email, push, SMS, WhatsApp)

PLATFORM OWNERSHIP

Primary
Adobe Analytics/GA4
Secondary
Contentsquare
Tertiary
WebEngage
Additional
BigQuery + Looker

EXTENDED COVERAGE VIA

Contentsquare does not perform marketing attribution natively — Adobe Analytics/GA4 is the right tool for that job. Cross-device attribution is deliberately routed through Contentsquare's own identity layer for greater accuracy on IndiGo's owned digital properties than a third-party ad-platform device graph provides.



08 Audience Management & Customer Segmentation

Compliant — combined stack

APPROACH

- WebEngage as the primary activation platform — dynamic segments, rule-based/RFM/predictive segmentation
- Airline-specific audiences pre-configured: frequent domestic/international, business, family, high-ancillary, at-risk, IROPS-impacted, loyalty candidates
- Real-time qualification during active booking sessions
- Direct activation across email, push, SMS, WhatsApp, in-app
- Contentsquare adds rich experience-based segments (frustrated users, error-impacted) that sharpen targeting precision
- GA4 audiences for Google Ads / DV360 targeting

PLATFORM OWNERSHIP

Primary
WebEngage
Secondary
Contentsquare
Tertiary
GA4
Additional
CDP

EXTENDED COVERAGE VIA

WebEngage orchestrates audience activation across every channel, with Contentsquare's behavioural intelligence sharpening targeting. Look-alike audiences are delivered via GA4 (paid media) and custom BigQuery ML (cross-channel).

AI-powered Experience analytics, Session replays & heatmaps

✓ 09 AI-Powered Analytics & Predictive Intelligence

Compliant — combined stack

APPROACH

- Adobe Analytics/GA4 as the primary predictive analytics engine: purchase probability, churn probability, predicted revenue
- Contentsquare Sense AI as secondary: automated root-cause analysis, NL queries (Sense Chat), AI-generated session/heatmap summaries, anomaly detection — diagnostic and generative, explaining why a metric moved
- WebEngage predictive segmentation: likelihood to purchase, optimal send time, channel affinity
- Custom ML models on BigQuery ML / Vertex AI for advanced predictions (CLV, demand forecasting, next-best-action)

PLATFORM OWNERSHIP

Primary
Adobe Analytics/GA4

Secondary
Contentsquare Sense AI

Tertiary
WebEngage

Additional
BigQuery ML / Vertex AI

EXTENDED COVERAGE VIA

GA4 is the primary engine for predictive scoring (propensity, churn, revenue prediction); Contentsquare Sense AI complements this with diagnostic and generative AI, extended with custom BigQuery ML / Vertex AI models for advanced use cases such as demand forecasting and next-best-action.

✓ 10–11 Digital Experience Analytics & Heatmaps

Compliant — category leader

APPROACH

- Contentsquare's founding capability. Full DXA and heatmap suite:
- Session replay (web + mobile) with PII privacy masking
- Rage click, dead click, excessive scrolling, form abandonment detection
- Customer struggle score and frustration index
- JS error, API failure, page rendering failure capture
- Impact Quantification — revenue impact per technical issue and per UI zone
- Click, scroll, movement, attention heatmaps with AI-assisted UX recommendations
- Mobile heatmaps for iOS and Android native apps

PLATFORM OWNERSHIP

Primary
Contentsquare

Additional
existing APM/Observability platform

EXTENDED COVERAGE VIA

All 18 DXA requirements and all 16 heatmap requirements are natively fulfilled. existing APM/Observability platform extends this with enterprise-wide technical observability correlation.

Reporting, self-service & data governance

✓ 12

Reporting, Dashboards & Business Intelligence

Compliant

APPROACH

- BigQuery as the primary, unified reporting layer — Looker Studio / Power BI dashboards spanning all platforms from one warehouse, avoiding reconciliation across separate tools
- Contentsquare as secondary — experience dashboards that drill through to session replays and heatmaps, a depth a warehouse-only view cannot replicate
- Adobe Analytics/GA4 scheduled marketing reports; WebEngage campaign reports, both also feeding BigQuery
- Export to PDF, Excel, CSV; native BigQuery connectors for Power BI / Tableau

PLATFORM OWNERSHIP

Primary
BigQuery

Secondary
Contentsquare

Tertiary
Adobe Analytics/GA4 + WebEngage

Additional
Looker / Power BI

EXTENDED COVERAGE VIA

BigQuery is the primary, unified reporting layer across all platforms; Contentsquare remains the secondary, experience-specific surface for session-level drill-through. Looker/Power BI Embedded extends native dashboards into IndiGo's internal applications where needed.

✓ 13-14

Self-Service Analytics & Data Quality Governance

Compliant

APPROACH

- All 3 platforms prioritise self-service: CS Sense Chat (NL queries), drag-and-drop dashboards; GA4 Explorations; WebEngage visual segment builder
- CDP Tracking Plans (schema registry) for schema validation
- GTM for tag deployment governance with version control
- CSQ PA SDK health monitoring for broken tracking detection
- Axeno will design Data Governance Playbook with stewardship roles

PLATFORM OWNERSHIP

Primary
Contentsquare

Secondary
GA4 + WebEngage

Additional
CDP

GTM

DataHub

EXTENDED COVERAGE VIA

Full enterprise data lineage and business glossary are delivered via DataHub, extending each platform's native governance capabilities into a unified enterprise view. Dashboard publishing is governed through role-based access control.

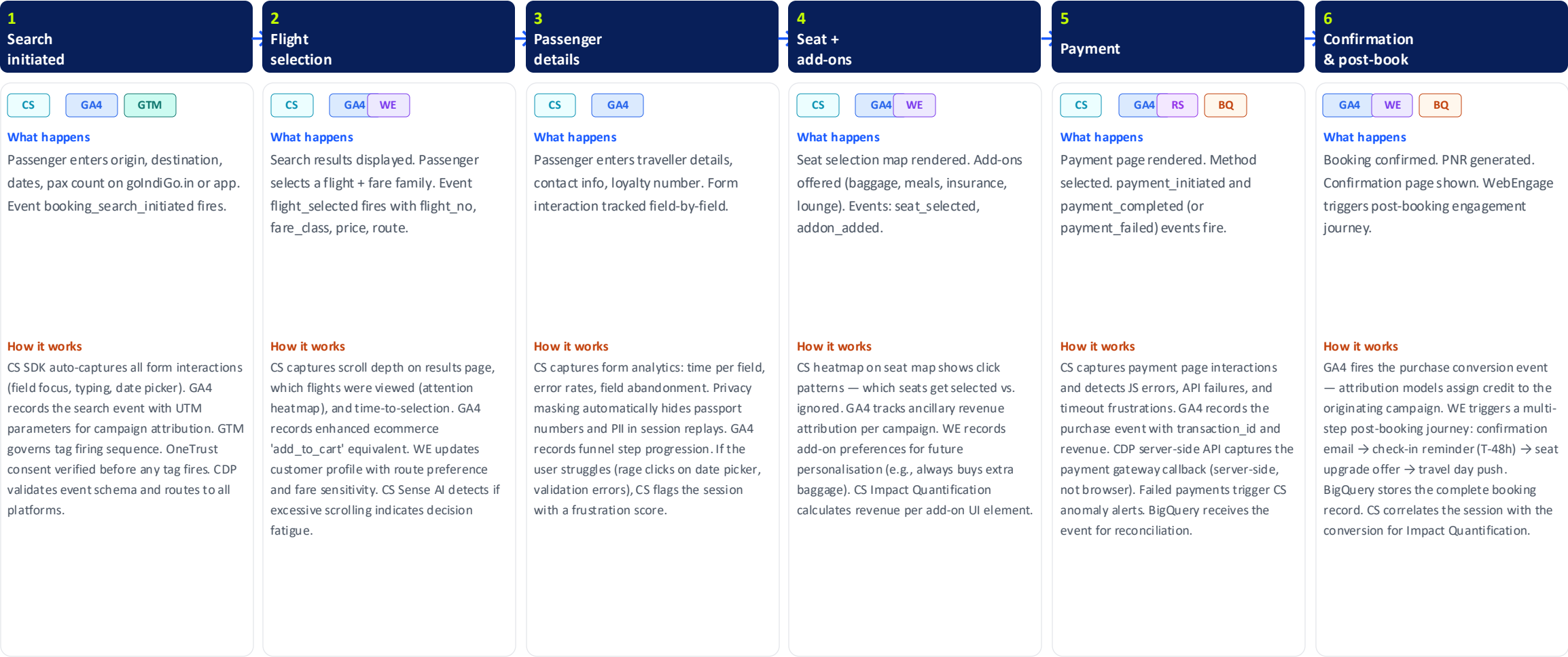
03

In-scope user journeys

Eight end-to-end journeys mapped step-by-step to the proposed architecture.
Each step shows what happens, which platforms are involved, and how it works.

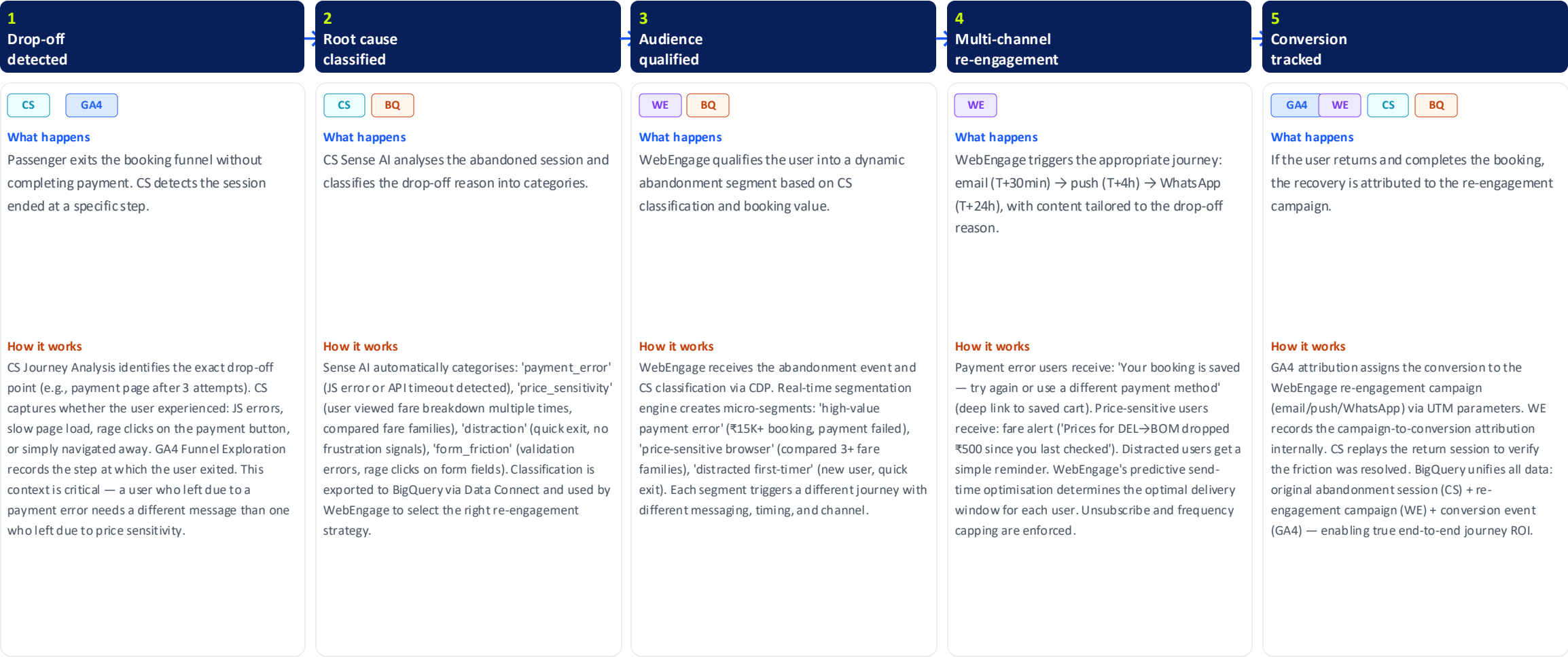
Journey 1 — Booking funnel (Search → Confirmation)

The primary revenue journey. Every step is instrumented, analysed for friction, and optimised through experimentation.



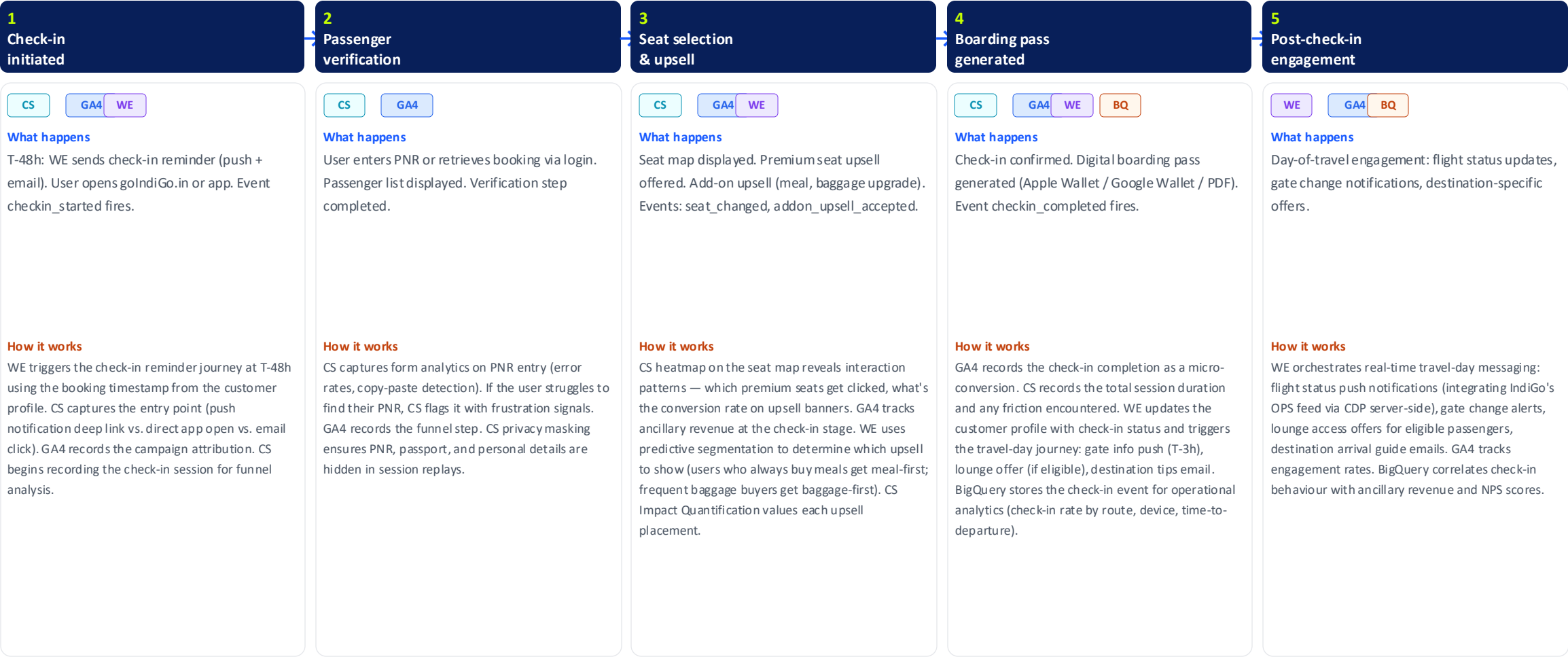
Journey 2 — Abandoned booking recovery

Detecting drop-offs with behavioural context and triggering intelligent re-engagement across channels.



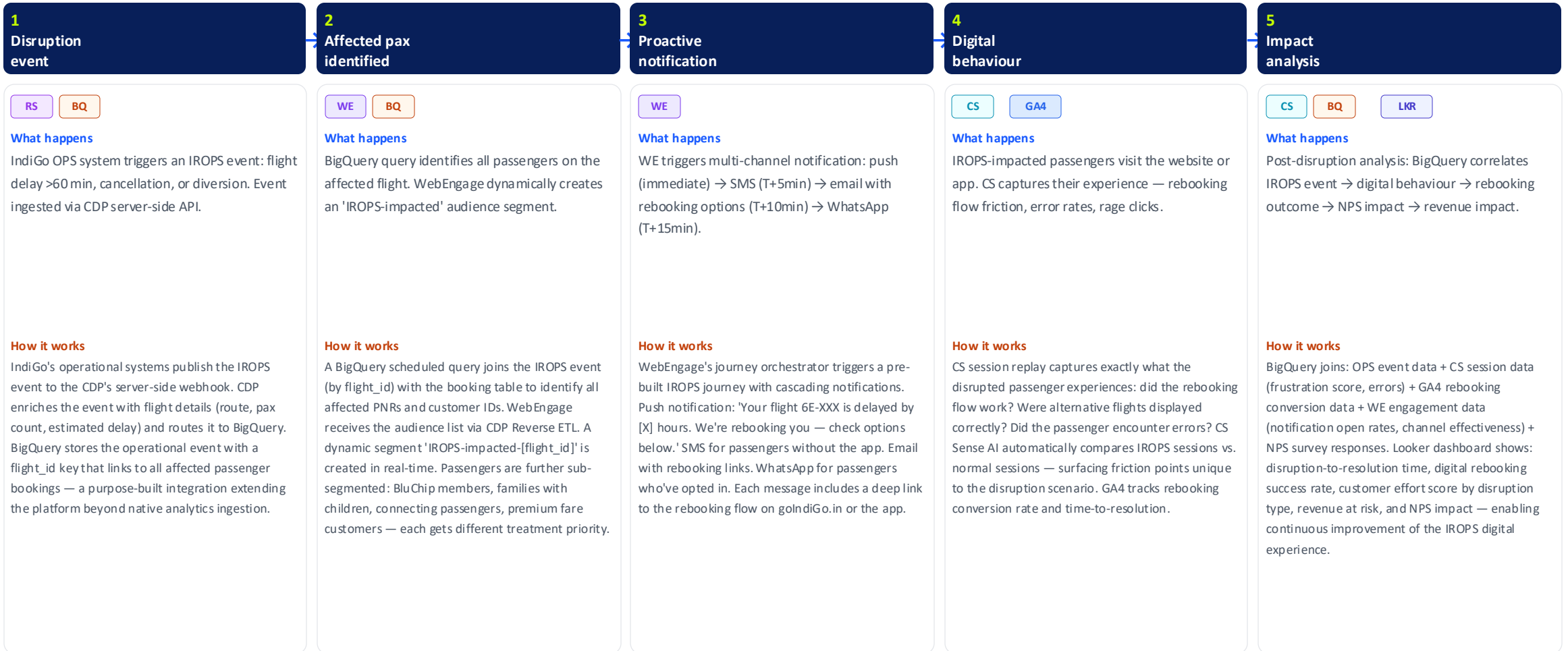
Journey 3 — Web & app check-in

From check-in initiation to boarding pass — optimising completion rates and ancillary upsell at every step.



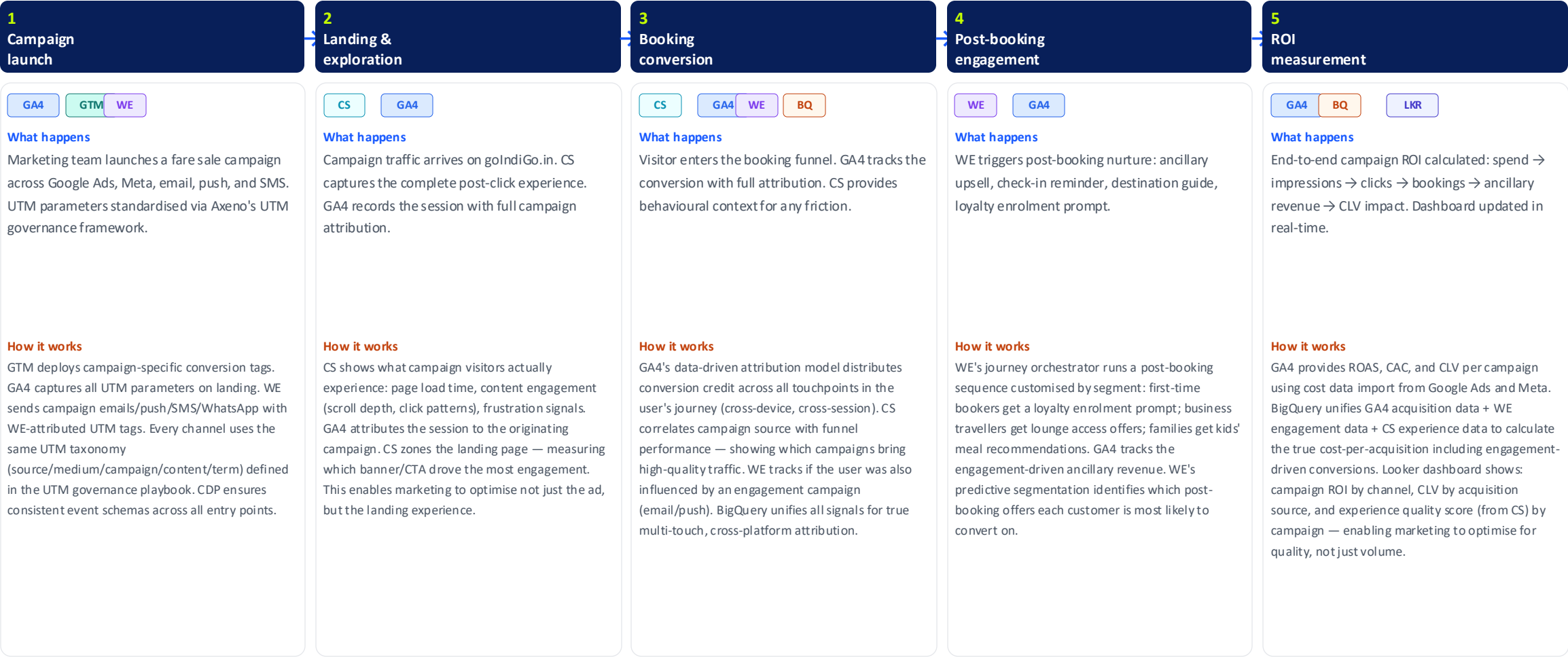
Journey 4 — IROPS & flight disruption management

When things go wrong — detecting operational events, measuring digital impact, and triggering proactive customer recovery.



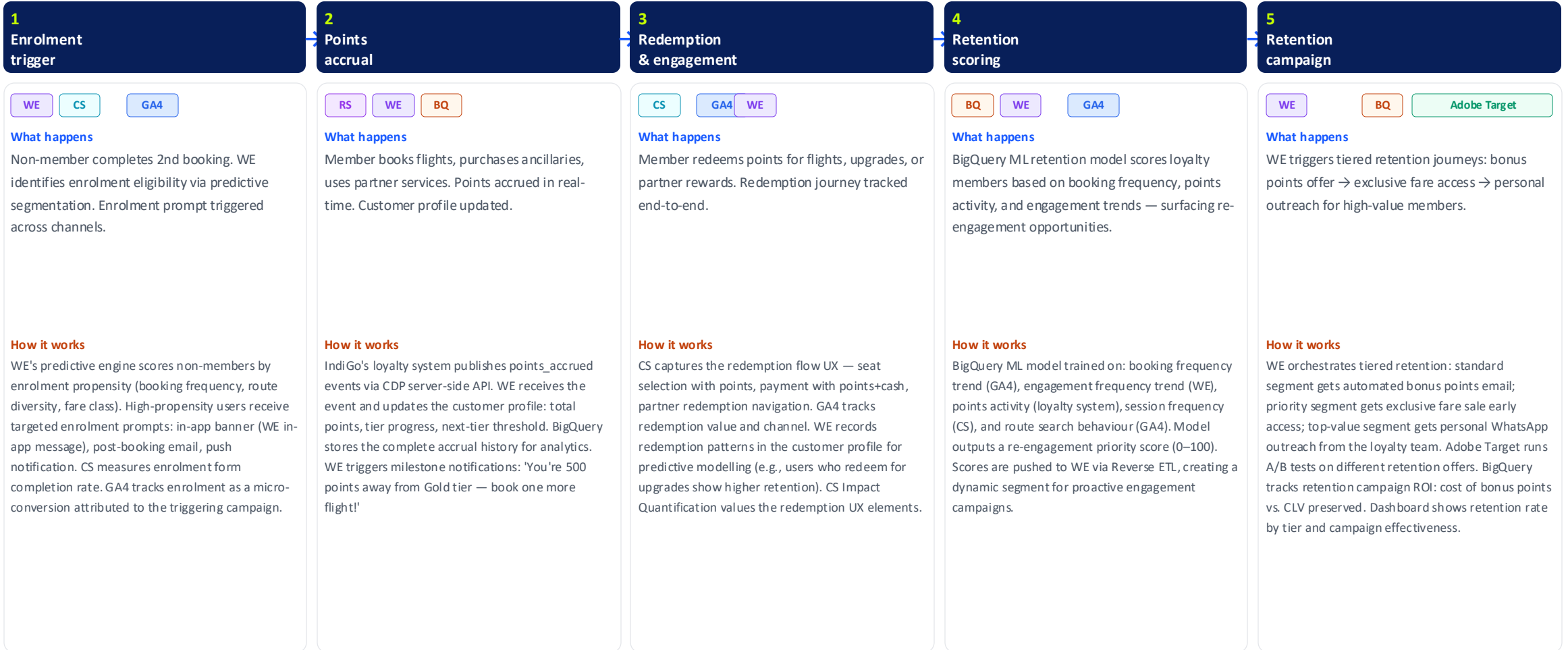
Journey 5 — Marketing campaign lifecycle (acquisition → conversion → retention)

Full-cycle campaign measurement from first impression through booking to repeat purchase, across paid, owned, and earned channels.



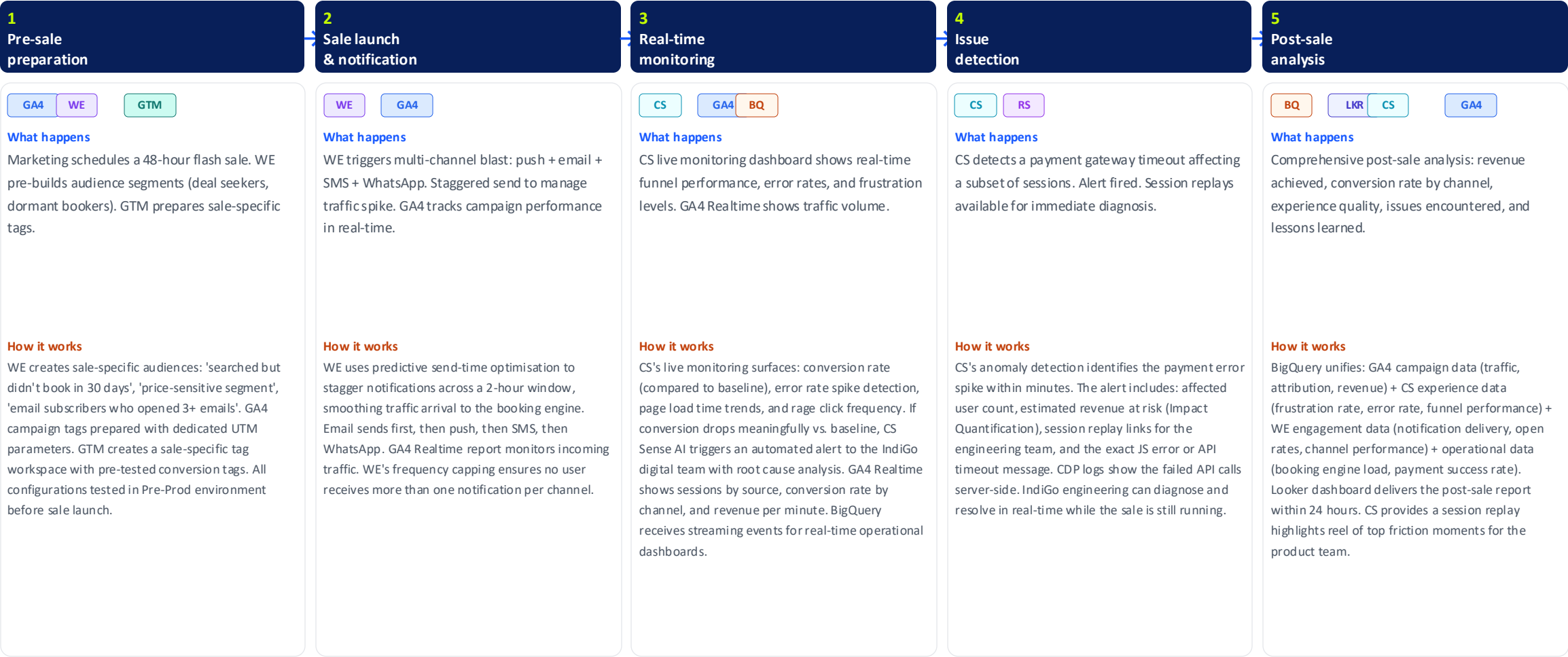
Journey 6 — BluChip loyalty programme lifecycle

From enrolment prompt to tier upgrade — tracking engagement, redemption patterns, and retention across the loyalty lifecycle.



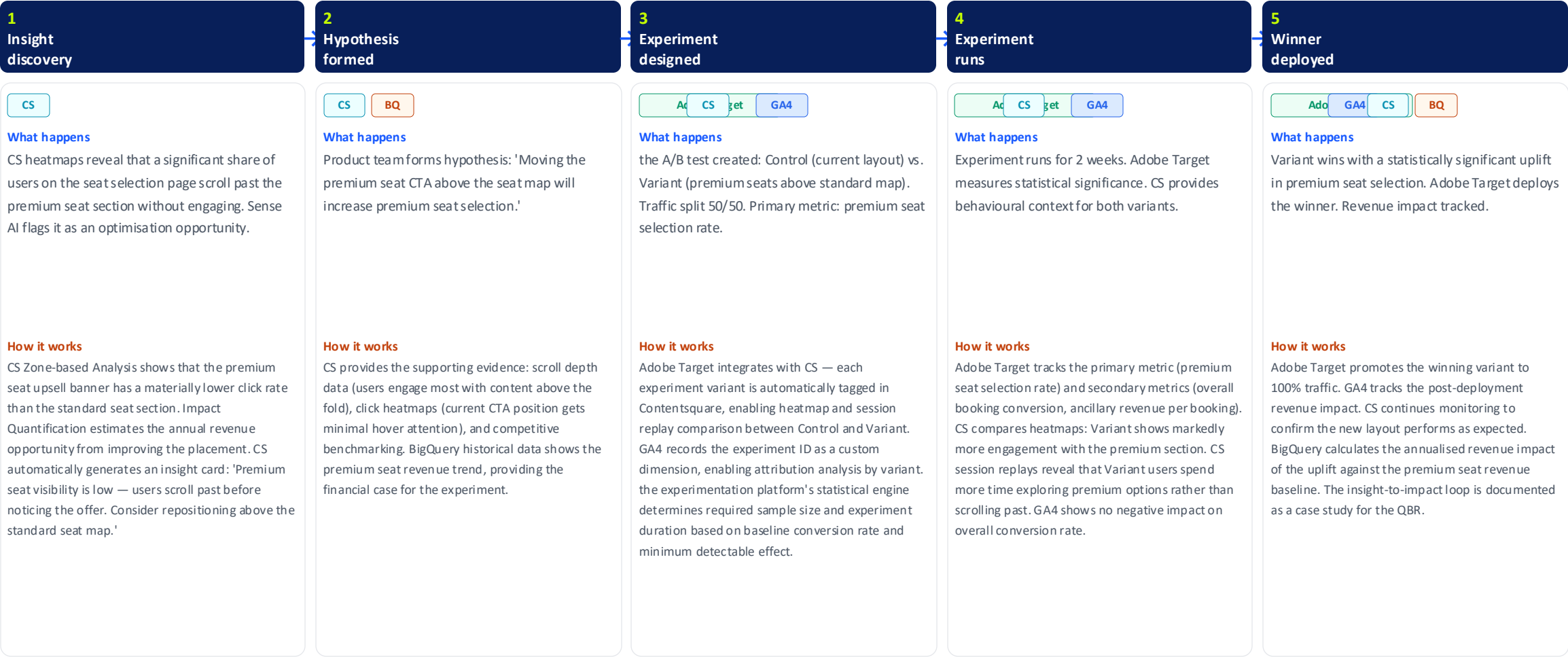
Journey 7 — Flash sale & peak traffic management

Real-time monitoring and optimisation during high-traffic events — fare sales, festival periods, and promotional launches.



Journey 8 — Experimentation & conversion rate optimisation (CRO)

From insight discovery (Contentsquare) to hypothesis, experiment design (Adobe Target), measurement (GA4), and rollout — the full optimisation loop.



04

Non-functional requirements

Architecture, security, performance, integration, SDK, and data management — the bar every workstream must clear.

Non-functional requirements — the bar every workstream must clear

1

Performance & scalability

15B+ sessions/day (CS) · horizontal auto-scaling everywhere · <3s dashboard latency

2

Architecture & design

Cloud-native, microservices, API-first · six environments (Dev→Prod) · HLA/LLA co-created with IndiGo

3

Security & compliance

ISO 27001/17/18/701, SOC 2 Type II · AES-256 + TLS 1.2+ · RBAC/MFA/SSO · PII auto-masked

4

Data residency

BigQuery Mumbai — the India-resident system of record for all three platforms, CSQ collects no PII data, DPDP-aligned

5

Integration & connectivity

120+ native integrations · REST/Webhooks/Kafka · Salesforce, BI tools, all major SDKs

6

SDK & implementation

<1% page-load impact · async, offline-queuing, kill-switch ·

7

Data retention

Configurable: 13mo (CS) · 14mo+ (GA4) · unlimited (BigQuery) — all India-resident

8

Support & managed services

24×7 Sev-1 (CS) + Delhi-based Axeno team · quarterly optimisation · hands-on training provided

Full detail, citations, and RFP requirement mapping for each area: Technical Appendix.

AI governance & responsible AI — Sense AI and predictive models

Classification & Scope

- Contentsquare Sense AI (Sense Chat/Sense Analyst), GA4 predictive metrics, and custom BigQuery ML/Vertex AI models are classified as Limited-risk AI — diagnostic and predictive analytics that inform human decisions, not automated decisioning affecting individual rights.
- No autonomous, credential-bearing AI agents are used anywhere in this architecture.

Human Oversight

- Every AI-assisted workflow keeps a human analyst in the loop: Sense AI insights and BigQuery ML predictions inform IndiGo's analytics team — no AI component acts autonomously on production systems.
- Anomaly detection alerts route to human review before any action is taken.

Data Use & IP Boundaries

- IndiGo data used in AI/ML components is used exclusively for IndiGo's contracted services — never mined for unrelated purposes or used to train general-purpose LLMs.
- All model outputs, dashboards, and configurations remain IndiGo-owned IP, consistent with the commercial terms in Section 8.

Incident Response & Change Control

- AI-related incidents (anomalous output, data quality issues affecting predictions) follow the same incident process as the rest of the platform — notification within 15 minutes, per the SLA framework.
- Model/config changes require IndiGo sign-off, with rollback supported via BigQuery ML versioning and Git-based history.

05

Extended capability roadmap

Complementary products and custom-delivered capabilities that extend the core platform combination to cover every RFP requirement in full.

Complementary products extending core platform coverage

Capability area	Relevant requirements	Recommended product	Type
A/B testing & experimentation	Req 6.1.18, Capability 9	Adobe Target	Existing
Tag management & governance	Capabilities 1, 2, 14	Google Tag Manager	Free
Event pipeline & identity resolution	Capabilities 3, 8, 14	CDP	OSS / Commercial
Unified data warehouse	Capabilities 4, 7, 9, 12	BigQuery (GCP Mumbai)	GCP
Enterprise BI dashboards	Req 12.1.2, 12.1.12, 12.1.15	Looker Studio / Power BI	Free / Commercial
Consent management (DPDP primary; GDPR for EU routes)	NF3 (Req 2.2.1–2.2.3)	OneTrust	Commercial
Technical observability	Req 10.1.17	existing APM / Observability platform	Commercial / OSS
Data catalogue & lineage	Req 14.1.5, 14.1.7, 14.1.13	DataHub	Open-Source
Custom ML models	Req 9.1.13, 9.1.14	BigQuery ML / Vertex AI	GCP
India data residency store	NF3 (data residency)	BigQuery Mumbai (asia-south1)	GCP

Custom-delivered capabilities — designed and built by Axeno

01 IROPS & operational event correlation

Correlating customer journeys with flight delays, fare changes, and schedule modifications. IndiGo OPS feed into BigQuery, with custom ETL joining journey data by timestamp and booking ID.

02 Household / family relationship modelling

Custom attribute design in WebEngage using PNR-derived family group IDs — extending the CDP with airline-specific relationship modelling.

03 Customer profile audit trail

Complete audit logging of all profile changes (merges, attribute updates, deletions) via a pipeline from CDP to a BigQuery audit table.

04 Advanced predictive ML models

CLV prediction, churn scoring, demand forecasting, next-best-action recommendation via BigQuery ML / Vertex AI pipeline design, training, and deployment. Phase 2 scope.

05 Cross-platform executive dashboards

Looker / Power BI dashboards combining Contentsquare + GA4 + WebEngage data via BigQuery. Unified KPI scorecards for CXO reporting.

06 UTM governance framework

Campaign taxonomy design, naming conventions, validation rules, and UTM builder tool for IndiGo's marketing team. Axeno deliverable during discovery.

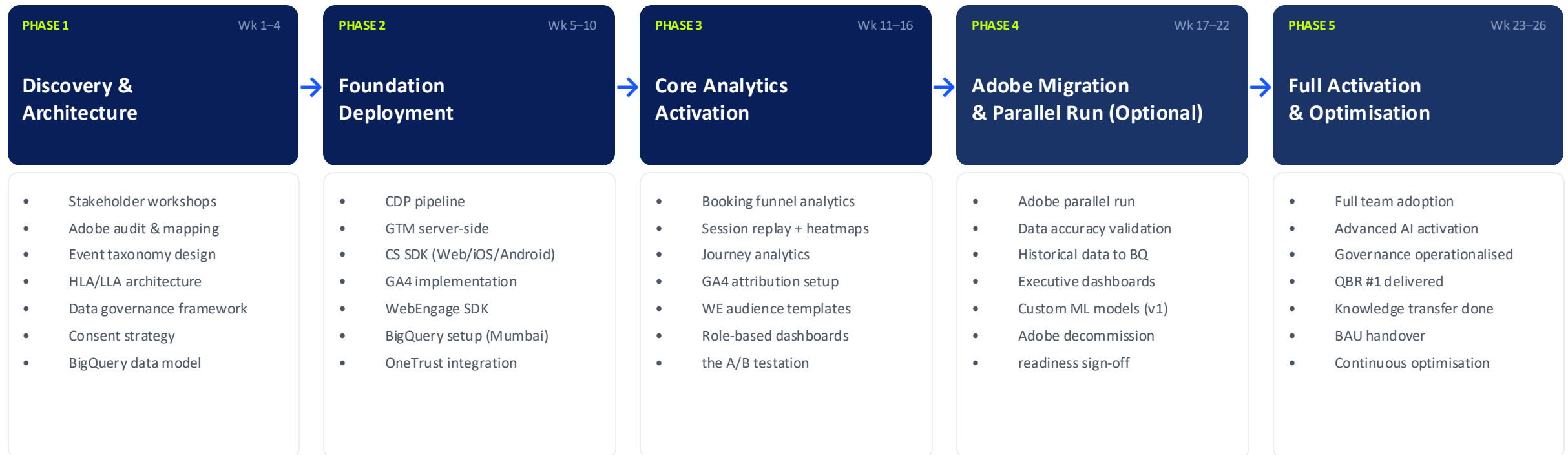
06

Implementation approach & timeline

A 26-week phased roadmap covering discovery, SDK deployment, analytics activation, Adobe migration, and full production readiness — with clear milestones at every gate.

Five-phase delivery methodology — discovery to optimisation

Our implementation follows Axeno's proven DDM (Discover–Design–Migrate) methodology, adapted for multi-platform analytics deployments. Each phase ends with a formal quality gate — IndiGo's sign-off is required before the next phase begins. The methodology is aligned with the RFP's requirements for phased rollout (Req 6.2.4), pilot deployments (Req 6.2.5), implementation best practices (Req 6.2.10), and knowledge transfer (Req 6.2.9).



Discovery & architecture — building the blueprint

Stakeholder workshops

- Executive sponsors: VP Digital, VP Marketing, VP Technology, VP CX
- Product owners for Web, App, Check-in, Loyalty, Ancillary
- Data & analytics team (current Adobe administrators)
- IT/security/compliance teams
- Contact centre & operations leads

Adobe Analytics audit

- Complete Adobe report suite inventory
- Event-to-event mapping (Adobe → CS + GA4 + WE)- if required, can be picked up with additional efforts
- Custom variables, eVars, props → GA4 dimensions/metrics
- Segment migration planning
- Scheduled report catalogue
- Data Feed export analysis for BigQuery historical load

Architecture & governance

- High-Level Architecture (HLA) document
- Low-Level Architecture (LLA) document
- Event taxonomy design (object_action naming standard)
- Data governance playbook
- BigQuery data model (star schema)
- CDP pipeline architecture
- Consent management strategy (OneTrust integration)
- Environment strategy (Dev/QA/UAT/PreProd/DR/Prod)

QUALITY GATE 1: Architecture sign-off · Event taxonomy approved · Migration mapping validated (if required) · Environment strategy confirmed

Foundation deployment — all SDKs live, all pipelines flowing

SDK deployment

- Contentsquare SDK on IndiGo website (desktop + mobile web)
- Contentsquare SDK on iOS app (Swift)
- Contentsquare SDK on Android app (Kotlin)
- GA4 gtag.js on website + Firebase SDK on iOS/Android
- WebEngage SDK on website + iOS + Android
- All SDKs deployed via GTM where applicable
- SDK health monitoring and diagnostics activated
- Privacy masking configured for PII fields

Pipeline & infrastructure

- CDP event pipeline deployment (server-side on GCP)
- GTM Server-Side Tagging container on GCP
- CDP Tracking Plans (schema registry) configured (schema validation)
- BigQuery dataset creation on GCP Mumbai (asia-south1)
- GA4 BigQuery Linking activated
- Contentsquare Data Connect configured
- WebEngage data export API connected to BigQuery
- OneTrust consent integration with all SDKs

Core event taxonomy

- Booking funnel events: search_initiated, flight_selected, passenger_details_entered, seat_selected, addon_added, payment_initiated, payment_completed, booking_confirmed
- Servicing events: checkin_started, checkin_completed, boarding_pass_generated
- Loyalty events: loyalty_enrolled, points_redeemed
- Engagement events: push_received, email_opened, sms_clicked
- All events validated via GA4 DebugView + CDP Live Events

QUALITY GATE 2: All SDKs live in production · Event data flowing to all platforms · BigQuery receiving cross-platform exports · SDK health green

Core analytics activation — from data to decisions

Experience analytics (CS)

- Booking funnel analysis — baseline benchmarks established
- Session replay activated with PII masking verified
- Heatmaps activated across all key pages
- Journey analysis configured (Search → Confirmation)
- Frustration detection (rage clicks, dead clicks, errors)
- Impact Quantification — revenue per element calibrated
- Sense AI onboarding for analyst team
- Anomaly alerting configured for key KPIs

Marketing & engagement

- GA4 attribution models configured (all 6 models)
- Campaign taxonomy and UTM governance framework deployed
- GA4 audiences created for Google Ads / DV360 activation
- WebEngage audience templates — 12+ airline-specific segments
- WebEngage journey orchestration (abandoned booking, IROPS)
- Cross-channel engagement flows (push + email + SMS + WhatsApp)
- WebEngage predictive segmentation activated
- Adobe Target (or Axeno-recommended alternative) integrated with CS + GA4

Dashboards & reporting

- Executive KPI dashboard (Looker Studio on BigQuery)
- Product analytics dashboard (CSQ PA)
- Marketing performance dashboard (GA4 + Looker)
- Engagement & retention dashboard (WebEngage)
- Booking funnel conversion dashboard (CS + GA4)
- Scheduled report distribution configured
- Role-based access: Executive, Product, Marketing, Ops, Tech
- Custom metrics & calculated dimensions deployed

QUALITY GATE 3: All 14 capabilities live · Dashboards reviewed by stakeholders · Baseline metrics established · Adobe Target first experiment live

Adobe migration (optional), parallel run, full activation & BAU handover

Phase 4: Adobe migration & parallel run (Wk 17–22)

- Adobe Analytics runs in parallel with CS + GA4 + WE for 6 weeks
- Data accuracy benchmarking — KPI-by-KPI comparison against Adobe baselines
- Historical data migration: Adobe Data Feeds → BigQuery (12-month lookback)
- Cross-platform executive dashboards built on Looker/Power BI (via BigQuery)
- Custom ML model v1 deployed on BigQuery ML: churn prediction + CLV scoring
- Adobe decommission readiness assessment: report parity checklist signed off
- All GA4 marketing attribution models validated against Adobe equivalents
- All Adobe scheduled reports migrated to GA4/Looker equivalents
- Adobe workspace-to-CS/GA4 mapping document delivered to IndiGo analytics team
- Formal Adobe decommission approval from IndiGo stakeholders

Phase 5: Full activation & optimisation (Wk 23–26)

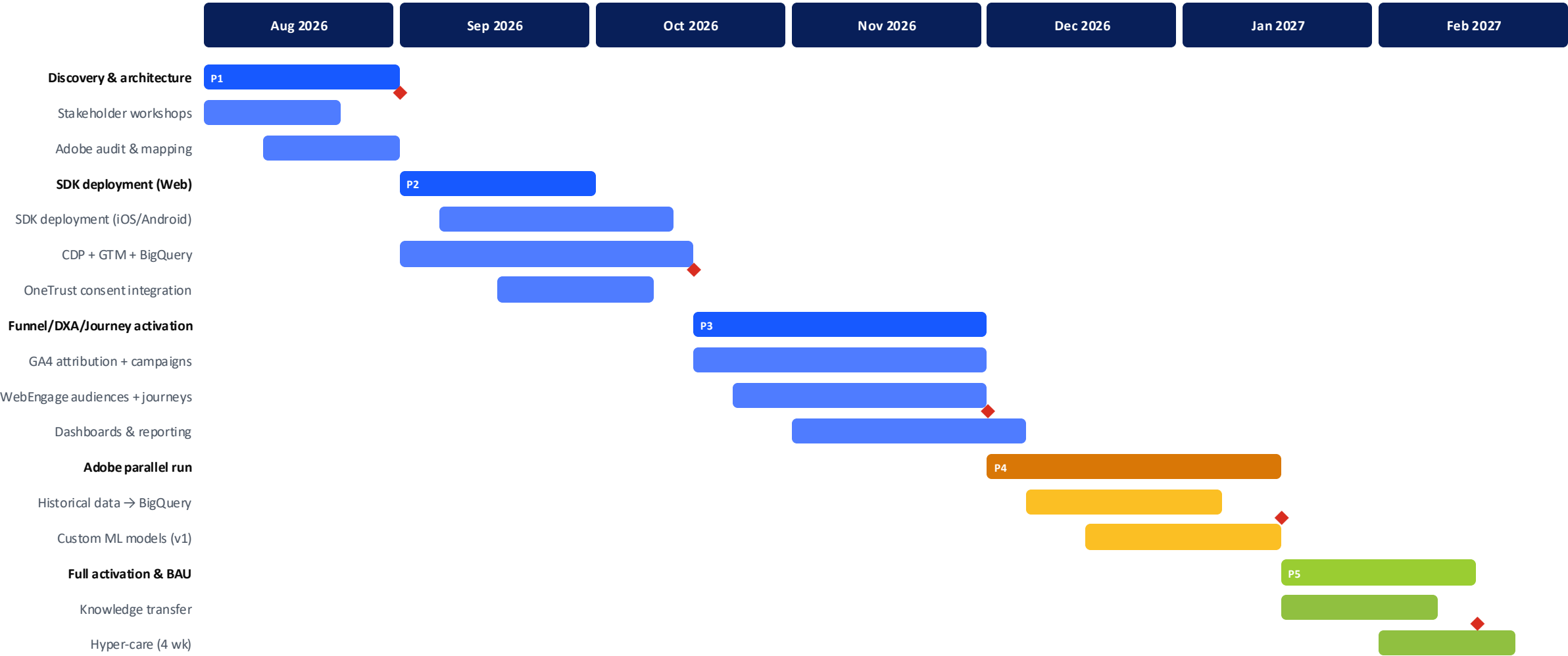
- Full production cutover — Adobe Analytics tags removed
- All IndiGo teams fully onboarded across all 3 platforms
- Advanced AI activation: predictive audiences, AI segments
- Data governance framework fully operationalised
- First Quarterly Business Review (QBR) delivered
- Knowledge transfer to IndiGo — admin, developer, business user tracks
- Analytics maturity assessment baseline established
- Continuous optimisation programme launched
- BAU managed services handover — SLA-based support begins
- Post go-live stabilisation support (4 weeks hyper-care)

GATE 4: Adobe decommission approved · KPI parity validated **GATE 5: Full production sign-off · BAU transition complete · KT signed off**

KEY IMPLEMENTATION DELIVERABLES

HLA & LLA documents · Event taxonomy playbook · Data governance framework · Adobe migration mapping · SDK implementation guides · BigQuery data model · Dashboard specifications · Training materials (admin/dev/business) · Environment runbook · SLA framework · QBR template · Analytics maturity scorecard

26-week Gantt — key activities and milestones



Six-environment strategy — Dev through Production

Environment	Purpose	Platform instances	Access
Development	SDK integration testing, event taxonomy validation, new feature development	CS sandbox, GA4 Dev property, WE test environment, BigQuery Dev dataset	Engineering team only
QA	Automated test execution, regression testing, event payload validation, SDK diagnostics	CS QA project, GA4 QA property, WE QA environment, BigQuery QA dataset	QA team + Engineering
UAT	Business user acceptance, dashboard review, report validation, stakeholder sign-off	CS UAT project, GA4 UAT property, WE UAT environment, BigQuery UAT dataset	Business users + Product owners
Pre-Production	Performance testing, load testing, production-parity validation, security testing	CS Pre-prod project, GA4 Pre-prod property, WE staging, BigQuery staging	DevOps + Security team
DR	Disaster recovery readiness, failover testing, data recovery validation	BigQuery cross-region replication, Cloud SQL point-in-time recovery + standby replica, Memorystore rebuilds from Cloud SQL (no DR needed — it's a cache, not a system of record), platform SaaS HA (vendor-managed)	DevOps + Platform admin
Production	Live customer data, real-time analytics, all dashboards & reports, audience activation	CS Production, GA4 Production, WE Production, BigQuery Production (Mumbai)	All roles (RBAC governed)
Promotion path: Dev → QA → UAT → Pre-Prod → Production · Configuration-as-code managed through GTM version control and CDP config management · Rollback supported at every stage (Req 5.1.11)			

Testing, quality assurance & defect management

Test Types

- Functional testing — event/dashboard validation against Adobe baseline
- Integration testing — CDP pipeline, API connections, cross-platform routing
- Load testing — executed in Pre-Prod ahead of flash sale / peak periods
- Security testing — SAST/DAST for all custom-built code (CDP transformations, BigQuery ETL, custom APIs)
- User Acceptance Testing (UAT) — business stakeholder sign-off before Production promotion

Test Environments & Gates

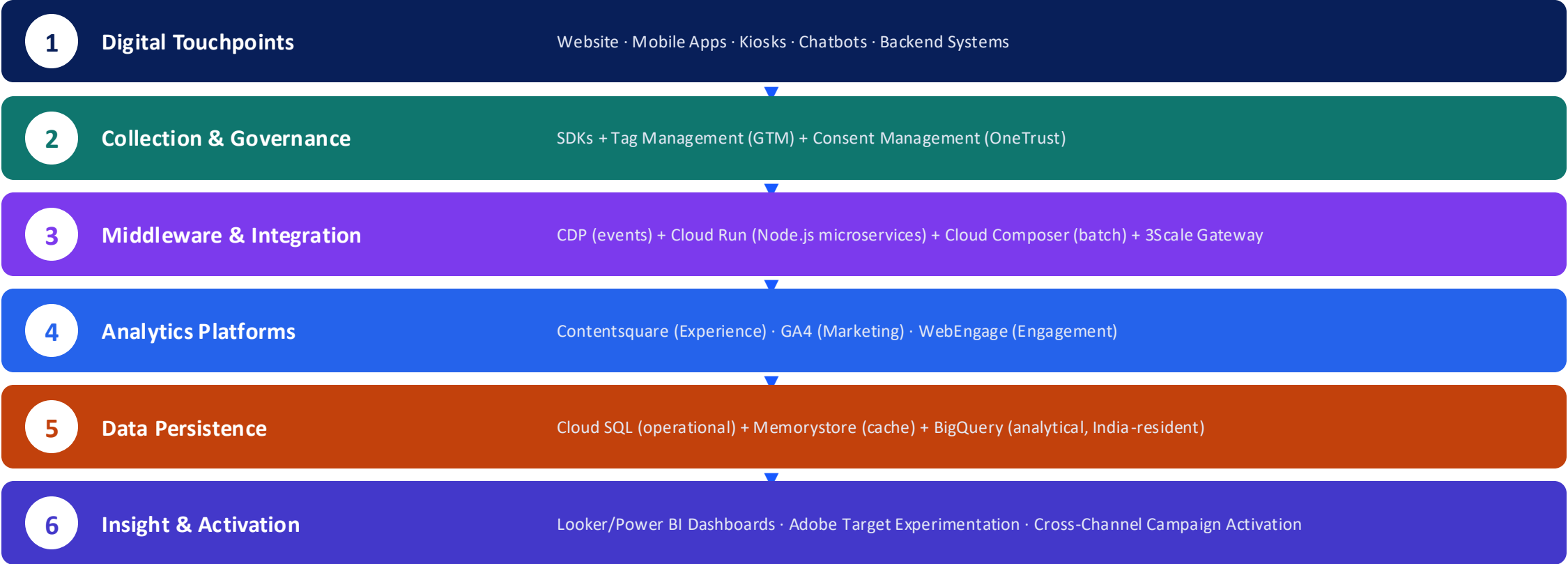
- Testing executed across Dev → QA → UAT → Pre-Prod, per the six-environment strategy (previous slide)
- Each of the 5 delivery phases closes with a formal Quality Gate (Section 6) requiring explicit IndiGo sign-off
- Adobe-vs-new-stack KPI benchmarking (deviation <2%) is a mandatory gate before Adobe decommission (Phase 4)

Defect Tracking & Access

- All defects are logged, triaged, and tracked in Axeno's project ticketing system (e.g., Jira) with defined severity SLAs
- IndiGo is granted read access to the defect tracking system for full transparency into status and resolution progress
- Post-Implementation Review (PIR) conducted after each major release to confirm security controls and business functionality remain intact

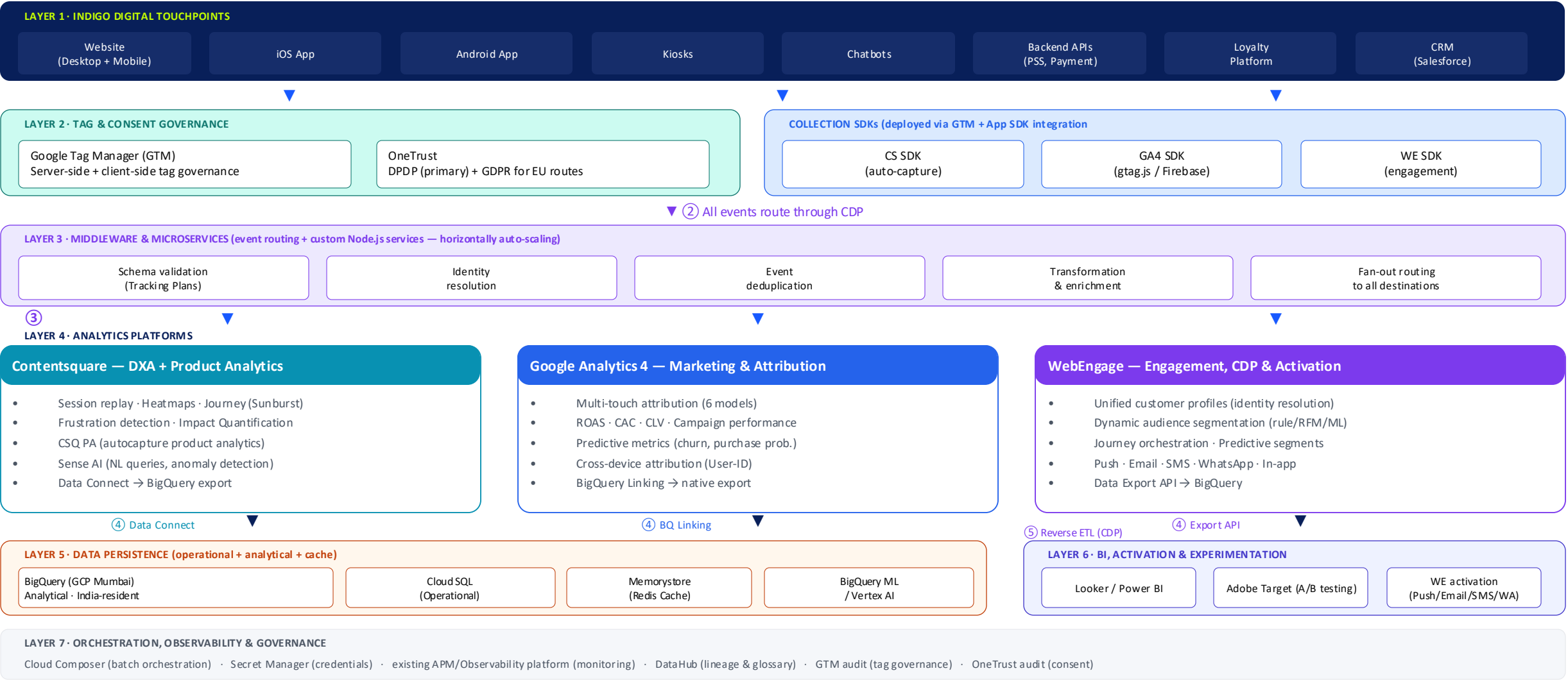
High-Level Architecture — six-layer executive view

A single-page summary for leadership review. The detailed component-level design (Low-Level Architecture) follows on the next page.



Governing principle: every layer is independently replaceable and independently scalable — the middleware layer in particular scales horizontally with event volume, decoupling every SaaS platform from every other and from the persistence layer beneath it.

Low-Level Architecture — seven-layer reference view with data flows



Data flows: ① Consent gates collection ② Events routed through CDP (middleware) ③ Fan-out to CS + GA4 + WE ④ Platform exports to persistence layer ⑤ Reverse ETL activates audiences ⑥ BI dashboards query BigQuery

Middleware architecture — scalability & resilience

CDP is the middleware layer of this architecture — every event from every source passes through it before reaching any destination. It is deliberately the one component every other layer depends on, so it is the one component engineered hardest for scale and resilience.

Horizontal Scalability

- Recommendation: the CDP/event-pipeline platform (SaaS), not self-hosted — it auto-scales transparently with event volume, needs zero GKE cluster management, and matches this architecture's serverless-first design throughout.
- Self-hosted CDP on GKE remains available only if IndiGo's data governance policy specifically requires it — priced and planned as an exception, not the default.
- Each data plane node is stateless; scaling out adds throughput linearly with no single-node bottleneck.

Resilience & Fault Tolerance

- At-least-once delivery guarantee with built-in retry and exponential backoff for every destination (Contentsquare, GA4, WebEngage, BigQuery) — a downstream outage queues events rather than dropping them.
- Dead-letter queuing captures any event that repeatedly fails delivery for manual review, so no event is silently lost.
- Multi-zone deployment on GCP ensures no single zone failure interrupts the pipeline.

Throughput & Latency

- Designed to comfortably absorb IndiGo's peak load: booking-funnel traffic during fare sales, IROPS-driven traffic spikes, and check-in-window surges — the same middleware layer that handles steady-state traffic handles the peak without a separate scaling event.
- Typical source-to-destination latency is sub-second for real-time destinations; batch destinations are scheduled independently (see Integration Patterns, next section).

Why middleware matters architecturally: without it, every SaaS platform would need a direct point-to-point integration with every data source — an $O(n^2)$ integration problem. CDP collapses this to a single hub, so adding a new source (e.g., a new mobile app) or a new destination (e.g., a future CDP) requires one new connection, not a rework of the entire pipeline.

Microservices layer (Node.js) — where it starts and where CDP ends

the CDP/event-pipeline layer's transformation logic is itself a lightweight Node.js execution environment, so it already covers most classic 'event glue code' — reshaping payloads, enrichment, conditional routing, deduplication. A separate microservices layer exists for the four jobs this CDP/event-pipeline layer is deliberately not built to do.

Already covered by CDP — no separate build needed

- Payload reshaping and field mapping between source and destination schemas
- Event enrichment from static or slowly-changing lookups
- Conditional filtering and routing rules
- Deduplication and schema validation against Tracking Plans
- Fan-out to multiple destinations from a single event

Needs the microservices layer (Cloud Run)

- Publishing custom REST APIs for IndiGo's own systems to call (fronted by 3Scale) — core scope
- Multi-step orchestration with state — the booking-confirmed saga touching loyalty accrual, CRM sync, and confirmation dispatch, with rollback on partial failure — core scope
- Custom webhook receivers needing business logic before an event even reaches CDP — core scope
- Real-time model serving (e.g., a future next-best-offer score) — the same layer extends to this if/when approved as a Year-2 initiative (Section 09), at incremental cost

Why Cloud Run, not a Kubernetes microservices fleet

Every other serverless component in this architecture (BigQuery, Vertex AI, Memorystore, the CDP/event-pipeline platform) scales to zero and needs no cluster management — a dedicated Kubernetes fleet just for a handful of custom functions would be the one inconsistent piece. Cloud Run and Cloud Functions run Node.js natively, scale to zero between calls, and require no persistent infrastructure to patch or capacity-plan. This is a small number of purpose-built functions, not a standing microservices platform — GKE remains reserved for the optional a self-hosted CDP/event-pipeline option path only, as already scoped in the DevOps/CI-CD slide.

Data persistence architecture — three purpose-built stores

Not all data belongs in the same store. This architecture uses three distinct persistence layers, each matched to a specific access pattern, rather than forcing everything into a single database.

BigQuery — Analytical Store

GCP Mumbai (asia-south1)

- Unified, India-resident system of record for all analytics data — booking events, session data, campaign data, engagement history
- Optimised for large-scale aggregation and ad-hoc SQL analysis, not single-record lookups
- Unlimited retention, columnar storage, serverless auto-scaling
- Feeds Looker/Power BI dashboards, BigQuery ML models, and Reverse ETL back to WebEngage

Cloud SQL — Operational Store

PostgreSQL, GCP Mumbai

- Transactional persistence for the middleware layer's own operational state: identity-resolution mapping tables, CDP configuration metadata, DataHub's metadata catalogue backing store
- Optimised for fast single-record reads/writes — the access pattern BigQuery is deliberately not optimised for
- Regularly backed up with point-in-time recovery, same India-resident region as BigQuery

Memorystore — Cache Layer

Redis, in-memory

- Sub-millisecond lookups for real-time operations: event deduplication windows, rate-limiting counters, session-state for in-flight identity resolution
- Reduces load on Cloud SQL and BigQuery for anything that needs to be checked on every single event
- Ephemeral by design — nothing in Memorystore is a system of record; it always has a durable source in Cloud SQL or BigQuery behind it

Data flow between the three: an incoming event is checked against Memorystore (dedup/rate-limit) → written to Cloud SQL if it updates operational state (e.g., an identity mapping) → streamed to BigQuery for analytical persistence. All three stores sit in GCP Mumbai, so India data residency applies uniformly across the full persistence layer, not just the analytical store.

Why each building block — the one-line version

Full justification, replaces/extends mapping, and data-flow detail for all 9 building blocks: Technical Appendix.

Contentsquare

Only platform combining session replay, heatmaps, and frustration detection in one stack

Google Analytics 4

Industry-standard attribution with native, zero-ETL BigQuery export

WebEngage

CDP-grade identity + full-channel activation — the piece nothing else in the stack does

CDP

Single middleware hub — eliminates the $O(n^2)$ point-to-point integration problem

Cloud Run

Serverless Node.js layer for custom APIs & orchestration CDP doesn't do

BigQuery (Mumbai)

India data residency by design + unifies all three platforms for cross-stack analytics

Cloud SQL + Memorystore

Operational + cache layer for the access patterns BigQuery isn't built for

GTM+OneTrust+Adobe Target+DataHub

Tag governance, consent, experimentation, and lineage — each a proven category leader

Cloud Composer+Secret Mgr

One scheduler for every batch job; one vault for every credential

End-to-end data journey — from passenger click to business decision

① Consent-gated collection

A passenger visits goIndiGo.in or opens the IndiGo app. OneTrust checks their consent preferences. If analytics consent is granted, GTM activates the Contentsquare, GA4, and WebEngage SDKs — ensuring collection only ever begins with explicit consent.

③ Server-side event capture

Backend events that don't originate in the browser — payment confirmations, booking engine events, PSS updates, loyalty point accruals, refund processing — are sent via the CDP's server-side API (or GA4's Measurement Protocol). These events carry the same canonical user_id, enabling cross-channel stitching.

⑤ Platform-level processing

Each platform processes events for its speciality. Contentsquare builds session replays, heatmaps, and journey maps. GA4 runs attribution models and calculates predictive metrics. WebEngage unifies customer profiles, qualifies audiences in real-time, and triggers engagement campaigns (abandoned booking flows, IROPS notifications, loyalty promotions).

⑦ BI dashboards & custom ML

Looker Studio and Power BI connect to BigQuery to render unified executive dashboards spanning experience (CS), marketing (GA4), and engagement (WE) data. BigQuery ML trains custom prediction models (CLV, churn, demand forecasting). Model outputs are pushed back to WebEngage via CDP Reverse ETL for audience activation.

② Client-side event capture

Contentsquare's Smart Capture SDK auto-records every interaction (clicks, scrolls, form inputs, rage clicks, dead clicks) without pre-instrumentation. GA4's gtag.js captures explicitly tagged marketing events (UTM parameters, campaign clicks, conversions). WebEngage's SDK captures engagement events (push token registration, in-app message views).

④ Event routing & governance

All events (client-side and server-side) flow through CDP. CDP validates every event against Tracking Plans (schema registry), performs identity resolution (merging anonymous IDs with authenticated user_ids), deduplicates events, applies transformations, and routes each event to its destination(s) — Contentsquare, GA4, WebEngage, and BigQuery simultaneously.

⑥ Data warehouse unification

All three platforms export processed data to BigQuery on GCP Mumbai (asia-south1): Contentsquare via Data Connect API, GA4 via native BigQuery Linking (zero-ETL), WebEngage via Data Export API. BigQuery becomes the single source of truth with India data residency — enabling cross-platform analytics beyond any single platform's native scope.

⑧ Audience activation & experimentation

WebEngage activates audiences across push, email, SMS, WhatsApp, and in-app channels. GA4 Audiences sync to Google Ads / DV360 for paid media targeting. Contentsquare experience segments (frustrated users, error-impacted) are exported to WebEngage for re-engagement. Adobe Target runs A/B tests informed by Contentsquare's funnel analysis.

DevOps, CI/CD & Infrastructure-as-Code

Infrastructure-as-Code

- All GCP infrastructure (BigQuery datasets, IAM policies, Vertex AI resources, GKE clusters where used) is defined and managed via Terraform — IndiGo's existing cloud-agnostic IaC standard.
- Terraform modules are delivered to IndiGo for full reproducibility and audit.
- Infrastructure changes follow a plan → review → apply pipeline with peer approval before any Production change.

Containerization & Orchestration

- The CDP/event-pipeline layer (SaaS option, fully managed) is the recommendation for this engagement — no container management required, no GKE cluster in the critical path.
- Self-hosted CDP on GKE is documented as a fallback only, priced separately if IndiGo's governance policy later requires it.
- Contentsquare, GA4, and WebEngage are vendor-managed SaaS and require no IndiGo-side containerization either.

CI/CD & Version Control

- GTM: workspace-based version control with preview/debug validation before every publish.
- CDP: Tracking Plans, transformations, and destination configs are managed as code in Git with automated testing gates.
- BigQuery SQL/ETL and Terraform configs: Git-based CI/CD with staged promotion across Dev → QA → UAT → Pre-Prod → Production (Section 6 Environment Strategy).

Feature toggles: GTM tags, CDP destinations, and WebEngage journeys can all be enabled/disabled instantly via configuration — no code redeployment required for standard changes.

API governance & integration standards

1

All Axeno-published APIs run as serverless Node.js functions on Cloud Run — RESTful and JSON-based (GraphQL considered only if IndiGo's ecosystem requires it)

2

Every custom API is published on IndiGo's 3Scale API Gateway, following IndiGo's standard API governance process

3

Token-based authentication (OAuth 2.0) is enforced on all published APIs, consistent with IndiGo's API security standards

4

TLS 1.2+ encryption in transit is enforced for every API call, with no exceptions

5

A complete, versioned API catalogue (OpenAPI/Swagger specification) is maintained and delivered to IndiGo

6

API-level caching is implemented at the BigQuery/Looker layer for dashboard-serving endpoints to optimise response times

7

Rate limiting and abuse protection are configured at the 3Scale Gateway layer for every published endpoint

8

External APIs (Salesforce CRM, payment gateway, loyalty platform) are assessed for security posture before integration, with access scoped via OAuth2 and rate limiting

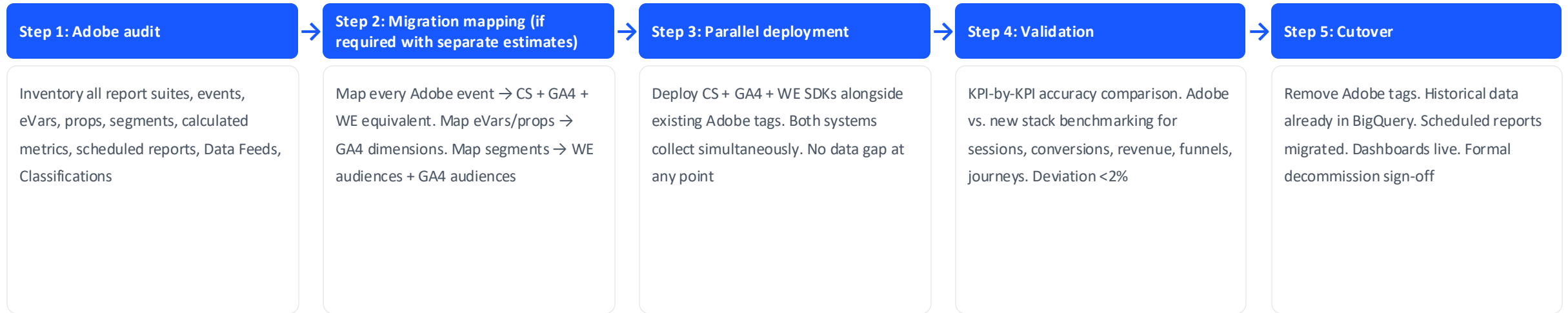
SCOPE NOTE: This is a Digital Analytics Platform proposal (Contentsquare, GA4, WebEngage). It does not include Passenger Service System (PSS) functions — IATA-standard Baggage, Cargo, or Teletype messaging are Not Applicable and remain with IndiGo's dedicated PSS/DCS vendors. File-based integration is not used anywhere in this design — all data exchange is API, SDK, or event-stream based.

07

Delivery confidence

Strategy, cutover plan, delivery governance, training programme, risk management, and the SLA framework that keeps everything on track.

Adobe Analytics to Contentsquare + GA4 — zero data continuity gap



Historical data migration scope

- Adobe Data Feeds (raw clickstream) → BigQuery: 12-month lookback minimum, enabling year-over-year comparisons from Day 1 of the new platform
- Adobe Classifications → BigQuery dimension tables: campaign taxonomies, product hierarchies, route classifications
- Adobe Segments → recreated as WebEngage audiences (behavioural) + GA4 audiences (marketing) + BigQuery SQL definitions (ad-hoc)
- Adobe Scheduled Reports → migrated to GA4 email reports + Looker Studio scheduled exports + CS anomaly alerts
- Adobe Workspace → dashboard equivalents built in CS (experience), GA4 (marketing), Looker (executive), WebEngage (engagement)
- Data validation protocol: Axeno runs automated KPI comparison scripts (BigQuery SQL) to detect deviations >2% across 50+ key metrics

Cutover strategy for a zero-disruption transition

Freeze windows

Agreed content and data freeze periods around cutover. No new Adobe implementations during parallel run. All stakeholders communicated well ahead.

Go / no-go criteria

Objective launch checklist: KPI parity validated (<2% deviation), all dashboards reviewed by business owners, SLA framework signed, rollback plan tested.

Rollback plan

Adobe tags remain deployable via GTM for 30 days post-cutover. Instant rollback by re-enabling Adobe tags and disabling new stack — no code push needed.

Hyper-care

4-week intensive post-cutover support. Dedicated Axeno team on standby. Daily data accuracy checks. Escalation path: Axeno PM → Axeno CTO → CS TAM.

DNS & redirects

No DNS changes required — analytics is tag-based, not URL-based. GTM handles all tag switching. Zero impact on website performance or availability.

Post-cutover validation

Weekly KPI comparison reports for 8 weeks post-cutover. Monthly accuracy audits for 3 months. Any deviation >2% triggers investigation and resolution.

Build-phase governance cadence

The Axeno delivery team (12 resources across architecture, platform implementation, microservices, data engineering, QA, and training — detailed in Section 8) operates on a structured governance rhythm from Day 1, ensuring IndiGo stakeholders have continuous visibility into progress, risk, and decisions.

DAILY	WEEKLY	BI-WEEKLY (weekly in P4)	QUARTERLY
- Stand-up: Axeno PM + IndiGo PM - SDK health check across CS, GA4, WE - Event validation report	- Sprint review: demos + blockers - KPI accuracy dashboard - Risk & issue log review	- Steering committee — escalated to weekly during Phase 4 (Adobe cutover, the highest-risk window) - Capacity & roadmap review - SLA performance report	- QBR: business value assessment - Analytics maturity scorecard - Optimisation recommendations - Product roadmap advisory

Full team composition, sizing rationale, and post go-live (BAU) team structure are detailed in Section 8 — Team, Support Track & Commercial Structure.

Knowledge transfer programme — three tracks, zero dependency

Administrator training

Platform admins, DevOps · 4 × 2-hour sessions

- User management & RBAC across CS, GA4, WE
- Environment configuration & promotion
- GTM workspace management & approval workflows
- CDP pipeline monitoring & schema management
- BigQuery access control & cost management
- SDK diagnostics & health monitoring
- OneTrust consent rule configuration
- Incident response & escalation procedures

Developer training

Frontend, backend, mobile engineers · 6 × 2-hour sessions

- CS SDK integration (Web, iOS, Android)
- GA4 gtag.js + Firebase SDK implementation
- WE SDK integration & event tracking
- Custom event implementation patterns
- Server-side event collection via Measurement Protocol
- CDP source/destination configuration
- BigQuery SQL for analytics queries
- SDK debugging & troubleshooting

Business user training

Product, marketing, analytics, CX teams · 8 × 90-min sessions

- Contentsquare: session replay, heatmaps, journeys, Sense AI
- GA4: Explorations, funnel analysis, attribution reports
- WebEngage: audience builder, journey designer, campaign analytics
- Looker/Power BI: executive dashboard navigation
- Self-service report creation across all platforms
- Alert & anomaly configuration
- A/B test design with Adobe Target
- Analytics-driven decision-making workshop

All training delivered on-site (Gurgaon) or remote · Materials in English & Hindi · Recorded sessions for future onboarding · Post-training competency assessment

Delivery risk register — proactive planning considerations

Consideration	Status	Context	Our approach
Data residency alignment	Managed	RFP requires India-resident hosting (Req 7.2.1) alongside global SaaS platform processing.	Addressed architecturally: BigQuery on GCP Mumbai (asia-south1) serves as the India-resident system of record, combined with Contentsquare's contractual DPDP framework and WebEngage's India-hosting option. Final legal sign-off from IndiGo's DPO recommended before contract signature.
Key-person / attrition risk	Monitored	Six of twelve build-phase roles are single-threaded specialists (Solution Architect, WebEngage Lead, Node.js Engineer, etc.) on a 6-month engagement — genuine attrition risk in the current market.	Named backup/shadow identified for each Tier-1 role from Axeno's bench before go-live; knowledge captured continuously (not just at handover) via shared documentation, pair-working in weeks 1-2 of each role, and recorded architecture walkthroughs — not fully eliminable, actively managed.
Adobe migration continuity	Managed	Historical report continuity during migration; KPI consistency between old and new platforms.	6-week parallel run with automated KPI comparison. Adobe Data Feeds backloaded to BigQuery for 12-month historical continuity. Deviation threshold: <2%, monitored daily. (can be picked up with additional efforts)
Multi-SDK performance	Managed	Coordinating CS, GA4, and WE SDKs alongside GTM requires careful load sequencing to preserve page performance.	All SDKs are async and non-blocking. GTM manages tag loading priority. Contentsquare publishes a <1% page-load impact benchmark in its own documentation (source available on request). Pre-launch load testing in Pre-Prod validates real-world performance on IndiGo's actual site.
Stakeholder adoption pace	Monitored	Business users transitioning from a single legacy tool to a three-platform ecosystem — adoption curves are inherently uncertain, not fully within our control.	Dedicated training programme (18 sessions across 3 tracks), on-site workshops in Gurgaon, Sense AI's natural-language interface to shorten the learning curve, and an adoption monitoring dashboard — residual risk depends partly on IndiGo's own change-management effort.
Discovery scope alignment	Monitored	Additional requirements may emerge during stakeholder workshops beyond the original RFP scope — common on multi-stakeholder engagements.	Fixed-scope discovery phase with formal change control process. Out-of-scope items documented transparently for Phase 2 backlog planning, with timeline impact assessed jointly before any scope addition is accepted.
Multi-platform coordination	Managed	Three specialised SaaS platforms require unified tooling to avoid operational fragmentation.	BigQuery serves as the vendor-neutral data layer. CDP provides platform-agnostic event routing. All data remains fully exportable and IndiGo-owned at every stage.

Service level commitments — aligned to RFP SLA criteria

SLA Metric	Target	Measurement	Platform
Platform availability	99.9%	Monthly uptime excluding scheduled maintenance windows	CS + GA4 + WE
Critical incident response	< 6 hours	Time from Sev-1 ticket to resolution (24x7 coverage)	Axeno + CS
Data processing latency (dashboard visibility)	< 30 min	Time from event collection to that event being reflected in an aggregated dashboard/report — distinct from raw event delivery, which is sub-2-second (see Integration Patterns, Architecture section)	CS + GA4
Dashboard availability	99.5%	Monthly uptime for all configured dashboards and reports	CS + GA4 + Looker
API availability	99.9%	Monthly uptime for all REST and event collection APIs	CS + GA4 + WE + RS
Data accuracy	≥ 99%	Monthly accuracy validation against baseline benchmarks	All platforms
Service request response	< 8 hours	Time from non-critical service request to response	Axeno
Security patch deployment	< 72 hours	Time from vendor security patch release to production deployment	Axeno + vendors
Incident notification	< 15 min	Time from incident detection to IndiGo notification	Axeno

SLA ALIGNMENT: Axeno commits to all SLA targets listed above, aligned with RFP Section VII service level criteria. Full SLA schedule, including any performance-linked commercial terms, will be finalised during contract negotiation.

CROSS-VENDOR TRIAGE: the APM platform's unified log correlation (all platforms + Cloud Run + BigQuery in one view) lets the on-call Axeno engineer isolate which layer an incident originates in within minutes, not hours — the 6-hour Sev-1 clock starts on detection, and root-cause isolation is the first 30 minutes of that window, not a separate open-ended investigation.

Axeno's organisational security & compliance posture

Beyond platform-level security (Contentsquare/GA4/WebEngage/GCP — detailed in the Platform Security & Compliance Pack), Axeno maintains its own organisational controls governing how our delivery team operates on IndiGo's engagement.

Access & Identity

- Documented Access Control Policy — provisioning, modification, revocation
- Unique user IDs, RBAC on least-privilege basis, enterprise MFA
- Privileged Access Management (PAM) for admin-level platform access
- Access revoked within 24 hours of personnel exit

Risk, Incident & Continuity

- Documented Risk Management Programme with periodic internal audits
- Information Security Incident Management with defined escalation and <15-minute IndiGo notification SLA
- Business Continuity / Disaster Recovery plan, tested annually
- Formal Change Management with security impact assessment on every change

Secure Delivery & Third Parties

- Secure SDLC for all custom-built components (CDP transformations, BigQuery ETL, Terraform)
- Background verification for all personnel deployed on IndiGo's engagement
- Third-Party Risk Management — subcontractors contractually bound to Axeno's security standards
- Mandatory annual security & privacy awareness training for all staff

Full control-by-control detail (129 individual security, privacy, and AI-governance controls spanning Access Management, Vulnerability Management, Physical Security, BCP/DR, and GenAI/Agentic AI governance) is provided in the separate RFP Questionnaire response and the Platform Security & Compliance Pack, both submitted alongside this proposal.

Assumptions, dependencies & open questions

Assumptions

- IndiGo will provide timely access to Adobe Analytics report suites, Data Feeds, and Classifications for migration audit
- IndiGo engineering teams will support SDK deployment on website and mobile apps within agreed sprint cycles
- GTM container access will be granted to Axeno with appropriate publishing permissions
- IndiGo will designate a dedicated Product Owner and Project Manager as counterparts to the Axeno team
- GCP Mumbai (asia-south1) billing account will be provisioned by IndiGo or jointly
- IndiGo's existing CRM (Salesforce), PSS, and loyalty platform APIs are documented and accessible for integration
- Platform licences (CS, WE, Adobe Target, OneTrust) will be procured by IndiGo directly with vendors, with Axeno facilitating negotiations
- No major website or app redesigns are planned during the 26-week implementation period

IndiGo dependencies — quantified

- Dedicated Product Owner: ~50% time for 26 weeks (single named counterpart, not a rotating pool)
- Dedicated Project Manager: ~50% time for 26 weeks
- VP-level stakeholders (Digital, Marketing, Technology, CX): ~2–3 hours/week each during Phase 1 (Wk 1–4), ~1 hour/week thereafter
- Engineering/IT: 1 named SDK-deployment contact per platform (web, iOS, Android), ~4–6 hours/week during Phase 2
- Business users: full-day availability for UAT sign-off at each Quality Gate (5 gates total)

Open questions for clarification round

- Confirming India data residency requirements to finalise the contractual vs. architectural compliance mix
- Are GraphQL/SOAP integrations actively used by IndiGo backend services?
- What is the current Adobe Analytics contract end date?
- Does IndiGo have an existing GCP or AWS cloud relationship?
- Is ABAC (Attribute-Based Access Control) a mandatory requirement?

08

Team, support track & commercial structure

Build-phase team, post go-live managed services, SLA commitments, commercial structure for implementation and ongoing support, and platform licensing.

Implementation team — 26-week build phase (Phase 1–5)

Role	#	Alloc.	FTE	Phase	Justification & responsibilities
Engagement Manager	1	50%	0.50	P1–P5	Escalation point, QBR, steering committee, commercial governance, IndiGo stakeholder alignment
Solution Architect	1	100%	1.00	P1–P3, 50% P4–5	HLA/LLA, event taxonomy, BigQuery data model, integration architecture, technical decision authority
CS Implementation Lead	1	100%	1.00	P1–P4, 50% P5	CS SDK (Web), DXA config, heatmaps, journey/funnel setup, Sense AI onboarding, Data Connect
CS Implementation Engineer	1	100%	1.00	P2–P5	CS SDK (iOS + Android), mobile heatmaps, app journey analytics, privacy masking config, Impact Quantification calibration
GA4 Implementation Lead	1	100%	1.00	P1–P4, 50% P5	GA4 property setup, gtag.js + Firebase SDK, attribution models, BigQuery Linking, Looker dashboards, UTM framework
WebEngage Lead	1	100%	1.00	P2–P5	WE architecture, customer profile schema, identity resolution config, journey orchestration design, predictive segmentation setup
WebEngage Specialist	1	100%	1.00	P2–P5	12+ airline audience templates, channel config (push/email/SMS/WhatsApp/in-app), IROPS + abandoned booking + loyalty journeys, campaign analytics
Senior Data Engineer	1	100%	1.00	P1–P5	CDP pipeline, GTM server-side, BigQuery ETL + star schema, Reverse ETL, OneTrust integration, schema governance
Data Engineer — Migration & ML	1	100%	1.00	P2–P5	Adobe Data Feed migration (12-month backload), BigQuery ML models (churn, CLV), Vertex AI pipeline, historical data validation
Node.js Microservices Engineer	1	100%	1.00	P3–P5	Custom Cloud Run APIs published on 3Scale Gateway, booking-confirmation orchestration (loyalty + CRM + confirmation webhook saga), custom partner webhook receivers — core scope, no Year-2 features
QA / Validation Lead	1	100%	2.00	P2–P5	Cross-platform event validation (CS + GA4 + WE + RS), SDK health monitoring, Adobe KPI benchmarking, go/nogo checklists
Training & Adoption Lead	1	50%	0.50	P3–P5	18-session programme (3 tracks), materials in English + Hindi, adoption dashboard, competency assessments, documentation

BUILD PHASE SUMMARY

12	Named resources	12.0	FTE equivalent	26	Week duration	5	Delivery phases	5	Quality gates
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Team sized for: 3 analytics platforms × 3 device surfaces (Web/iOS/Android) + CDP pipeline + Cloud Run microservices + BigQuery DWH + Adobe migration + 12month historical backload + BigQuery ML + 18 training sessions

Why 12 resources — mapping team to scope complexity

CS: 2 engineers (Lead + Engineer)

Contentsquare must be deployed across 3 surfaces: desktop web, iOS native, and Android native. Each surface has distinct SDK integration patterns, privacy masking rules, and heatmap configurations. The Lead handles web (highest traffic surface) + DXA strategy + Sense AI. The Engineer handles iOS + Android SDKs, mobile heatmaps, and app-specific journey analytics. Parallel deployment shortens Phase 2 by ~3 weeks.

WebEngage: 2 specialists (Lead + Specialist)

WebEngage spans 6 distinct modules requiring configuration: CDP/identity resolution, audience segmentation engine, journey orchestrator, channel setup (push + email + SMS + WhatsApp + in-app), predictive segmentation ML, and campaign analytics. The Lead designs the architecture. The Specialist builds 12+ airline audience templates, configures all 5 channels, and builds the IROPS/abandoned booking/loyalty journeys — a two-person team delivers this fully within the 5-month window.

Data Engineering: 2 engineers (Senior + Migration/ML)

The scope spans two parallel workstreams: (1) Infrastructure — CDP pipeline, GTM server-side container, BigQuery star schema design, Reverse ETL, OneTrust consent integration. (2) Migration & ML — Adobe Data Feed backload, BigQuery ML model development, Vertex AI pipeline, historical data validation. Running these concurrently in P2–P4 keeps the 26-week timeline intact.

Microservices: 1 engineer (Node.js / Cloud Run)

Core scope requires custom API publishing (any IndiGo-facing endpoint on 3Scale) and stateful orchestration (the booking-confirmation saga touching loyalty accrual, CRM sync, and confirmation dispatch, with rollback on partial failure) — this is application backend development, a distinct skillset from pipeline configuration. One dedicated engineer builds and hardens this layer across P3–P5, once the core pipeline and data model it depends on are stable. Note: this role is NOT what builds the Year-2 next-best-offer idea (Section 09) — that would be scoped and staffed separately if approved.

Post go-live managed services — scope, SLA, and governance

MANAGED SERVICES SCOPE

Platform operations

- SDK health monitoring & incident response across CS, GA4, WE
- Event taxonomy maintenance — new events, schema changes, deprecation
- GTM tag management — new tag deployment, version control, approval workflows
- CDP pipeline monitoring, schema validation, identity resolution tuning
- BigQuery data model maintenance, partition management, query optimisation

Analytics & optimisation

- Monthly analytics insights report — top friction points, conversion opportunities
- Quarterly Business Review (QBR) — business value assessment, maturity scorecard
- Dashboard maintenance — new metrics, dimension changes, new report requests
- A/B test support — hypothesis development, the A/B test setup, results analysis
- Sense AI insight curation — surfacing and prioritising AI-generated recommendations

Campaign & audience support

- WebEngage audience template creation and maintenance
- Engagement journey optimisation — abandoned booking, IROPS, loyalty, lifecycle
- GA4 audience management for Google Ads / DV360 activation
- Campaign attribution analysis and marketing ROI reporting

Governance & training

- Ongoing training for new team members (recorded sessions + live workshops)
- Data governance — compliance monitoring, consent audit, data quality scoring
- Platform upgrade advisory — CS, GA4, WE release notes, feature activation
- Quarterly analytics maturity assessment and roadmap recommendations

SLA COMMITMENTS

Severity	Target	Coverage
Sev-1 (Platform down)	< 6 hour response	24x7
Sev-2 (Feature degraded)	< 8 hour response	Business hours
Sev-3 (Non-critical issue)	< 2 business day response	Business hours
Service request	< 1 business day acknowledgement	Business hours
Scheduled changes	48-hour advance notification	Planned window
Monthly uptime	99.5% managed components	Monthly

GOVERNANCE CADENCE (POST GO-LIVE)

WEEKLY

Platform health report, open ticket review, upcoming change calendar

MONTHLY

Analytics insights report, SLA performance scorecard, capacity review, steering committee

QUARTERLY

QBR with business value assessment, analytics maturity scorecard, optimisation roadmap, product advisory

Post go-live team — scaled for operational complexity

BUILD (Wk 1–26)

12 resources · 11.0 FTE

▼ 4-week hyper-care transition ▼

BAU SUPPORT (Month 7+)

8 resources · 5.75 FTE

VENDOR SUPPORT (IN PLATFORM LICENCE)

- CS: Dedicated TAM + CSM, 24×7 Sev-1, quarterly platform reviews
- GA4: Google Cloud support (Premium), BigQuery SLA, Firebase support
- WE: Dedicated CSM, India timezone, SLA-based support, roadmap advisory

Role	#	Alloc.	FTE	Scope & justification
Engagement Manager	1	25%	0.25	QBR delivery, steering committee, escalation management, roadmap advisory, commercial oversight
DXA / Experience Analyst	1	100%	1.00	CS-focused: monthly friction analysis, session replay reviews, heatmap insights, Sense AI curation, conversion optimisation recommendations, the A/B test support
Marketing Analytics Analyst	1	100%	1.00	GA4-focused: attribution analysis, campaign performance reporting, ROAS/CAC/CLV monitoring, Looker dashboard maintenance, audience management for Google Ads/DV360
WebEngage Specialist	1	100%	1.00	Audience template maintenance, journey optimisation (abandoned booking, IROPS, loyalty, lifecycle), predictive segment tuning, channel performance, new campaign setup
Platform Administrator (CS + GA4 + GTM)	1	100%	1.00	SDK health monitoring, GTM tag management, event taxonomy maintenance, CS/GA4 platform upgrades, incident response, access management, new tag deployments
Platform Administrator (WE + CDP)	1	100%	1.00	WE platform ops, CDP pipeline monitoring, schema validation, identity resolution tuning, channel deliverability, integration health, incident response
Data Engineer	1	50%	0.50	BigQuery ETL maintenance, data model updates, ML model retraining (quarterly), Reverse ETL tuning, Cloud Run microservice maintenance (low-touch, serverless), new integration requests
QA Analyst	1	50%	0.50	Ongoing event validation, SDK health reports, data accuracy audits (monthly), regression testing for platform upgrades, dashboard accuracy verification

BAU TOTAL: 8 resources · 5.75 FTE · ~43% cost reduction vs. build · No added headcount for the microservices layer — Cloud Run is serverless and low-touch, absorbed within the existing Data Engineer role · Quarterly re-sizing, 30-day notice · Minimum floor: 4.0 FTE

Implementation commercials — 12 resources, 26 weeks

Fixed-price delivery; Software pricing to be sized based on session volume. Pricing will be provided when sizing data is shared

Role	#	Monthly rate (₹)	Alloc.	Months	Total (₹)
Engagement Manager	1	[Rate card]	50%	6	[Computed]
Solution Architect	1	[Rate card]	100%	6	[Computed]
CS Implementation Lead	1	[Rate card]	100%	6	[Computed]
CS Implementation Engineer	1	[Rate card]	100%	5	[Computed]
GA4 Implementation Lead	1	[Rate card]	100%	6	[Computed]
WebEngage Lead	1	[Rate card]	100%	5	[Computed]
WebEngage Specialist	1	[Rate card]	100%	5	[Computed]
Senior Data Engineer	1	[Rate card]	100%	6	[Computed]
Data Engineer — Migration & ML	1	[Rate card]	100%	5	[Computed]
Node.js Microservices Engineer	1	[Rate card]	100%	4	[Computed]
QA / Validation Lead	1	[Rate card]	100%	5	[Computed]
Training & Adoption Lead	1	[Rate card]	50%	4	[Computed]
TOTAL BUILD PHASE	12		11.0 FTE	26 wk	₹ [Total]

PAYMENT MILESTONES

- 10%

Contract signing
+ discovery kick-off
- 15%

Gate 1: Architecture
signed off (Wk 4)
- 20%

Gate 2: All SDKs live
in production (Wk 10)
- 20%

Gate 3: All capabilities
activated (Wk 16)
- 25%

Gate 4: Data accuracy
validated, Adobe decom. (Wk 22)
- 10%

Gate 5: Full go-live
& KT complete (Wk 26)

Managed services commercials — monthly retainer (post go-live) - TBC

Monthly retainer with committed capacity. Quarterly re-sizing option — scale for peak initiatives or steady-state. Minimum floor: 4.0 FTE.

Role	#	Monthly rate (₹)	Alloc.	Monthly cost (₹)	Scope covered
Engagement Manager	1	[Rate card]	25%	[Computed]	QBR, steering committee, escalation, roadmap advisory
DXA / Experience Analyst	1	[Rate card]	100%	[Computed]	CS insights, session replay, heatmaps, Sense AI, CRO, Adobe Target support
Marketing Analytics Analyst	1	[Rate card]	100%	[Computed]	GA4 attribution, campaign reporting, ROAS/CAC/CLV, Looker dashboards
WebEngage Specialist	1	[Rate card]	100%	[Computed]	Audiences, journeys, predictive segments, channel ops, campaigns
Platform Admin (CS+GA4+GTM)	1	[Rate card]	100%	[Computed]	SDK health, GTM tags, event taxonomy, CS/GA4 upgrades, incidents
Platform Admin (WE+RS)	1	[Rate card]	100%	[Computed]	WE platform ops, CDP pipeline, schema, identity, integrations
Data Engineer	1	[Rate card]	50%	[Computed]	BigQuery ETL, ML retraining, Reverse ETL, new integrations
QA Analyst	1	[Rate card]	50%	[Computed]	Event validation, SDK health reports, data accuracy audits
TOTAL MONTHLY RETAINER	8		5.75 FTE	₹ [Total] / mo	Full managed services scope

COMMERCIAL TERMS

12-month minimum, auto-renewing · Monthly in arrears, NET 30
Annual CPI revision capped at 5% · Quarterly re-sizing, 30-day notice
Minimum floor: 4.0 FTE · Exit: 90-day notice + full KT
All IP (configs, dashboards, models, docs) owned by IndiGo

OPTIONAL ADD-ONS (incremental per month)

- Dedicated ML Engineer (50%) — advanced BigQuery ML, Vertex AI
- Content / UX Strategist (50%) — CRO recommendations, test hypothesis
- Additional the A/B test capacity — beyond base allocation
- Peak surge capacity — short-term scale-up for flash sales, app launches

Platform licensing — indicative annual costs (Provided post sizing)

NOTE: Platform licences are procured by IndiGo directly with each vendor. Axeno facilitates negotiations and provides commercial benchmarking. Final pricing subject to vendor negotiation based on IndiGo's traffic volumes and feature requirements.

Platform	Licence model	Sizing basis	Indicative annual (USD)	Notes
Contentsquare (DXA + CSQ PA)	Annual subscription	Sessions / month (web + app combined)	[Quote provided post sizing data]	Includes DXA, Product Analytics, Sense AI, Data Connect. Pricing tier based on session volume.
Google Analytics 4	GA4 360 (enterprise)	Properties, events, BigQuery export	[Quote for 360 — treat as required]	At IndiGo's traffic volume (100M+ annual passengers), GA4 standard (free) will hit Google's sampling thresholds on ad-hoc analysis and carries no enterprise SLA. GA4 360 is the realistic requirement, not an optional upgrade — budget for it as a mandatory line item, not a nice-to-have.
WebEngage	Annual subscription	Monthly Active Users (MAUs)	[Quote provided post sizing data]	Includes CDP, journey orchestration, all channels (push, email, SMS, WhatsApp, in-app), predictive segmentation.
Adobe Target	Annual subscription	Monthly tested users	Existing vendor	Experimentation platform. Pricing based on tested visitor volume. Native CS + GA4 integration included.
OneTrust	Annual subscription	Domains + sessions + regulations	[Quote provided post sizing data]	Primary requirement is DPDP (India). GDPR support covers EU-origin passenger data on IndiGo international routes, not a separate compliance programme. Pricing based on domains, traffic volume, and regulatory frameworks.
BigQuery (GCP)	Pay-per-use	Storage (TB) + Queries (TB scanned)	[Quote provided post sizing data]	GCP Mumbai (asia-south1). Storage: ~\$0.02/GB/month. Queries: \$5/TB scanned. Flat-rate pricing available for predictable workloads.
CDP	Cloud (recommended)	Events / month	[Cloud quote]	Recommended: the CDP/event-pipeline platform — managed infrastructure, dedicated support, zero cluster management, consistent with this architecture's serverless-first design. Self-hosted OSS on GCP remains a zero-licence-cost fallback only if IndiGo's governance policy requires it.
DataHub	Open-source	Self-hosted on GCP	Free (OSS)	Open-source data catalogue. Hosted on GCP alongside BigQuery. No licence cost — only GCP compute costs.

Axeno will provide a detailed licensing estimation workbook during the clarification round, based on IndiGo's confirmed traffic volumes (monthly sessions, MAUs, events) and feature requirements.

Commercial structure at a glance

BUILD PHASE

Fixed price · 26 weeks

Team

12 resources / 11.0 FTE

Duration

26 weeks (6 months)

Model

Fixed-price, milestone-based

Payments

6 milestones tied to quality gates

Key roles

SA, 2× CS, GA4, 2× WE, 2× DE, Node.js Eng, QA, Training

Total

₹ [To be finalised]

SUPPORT TRACK

Monthly retainer · 12-month min

Team

8 resources / 5.75 FTE

Duration

12-month minimum, auto-renew

Model

Monthly retainer, quarterly re-sizing

SLA

99.5% uptime, <4h Sev-1 resolution

Key roles

DXA analyst, Marketing analyst, WE, 2× Admin, DE, QA

Monthly

₹ [To be finalised] / month

PLATFORM LICENSING

IndiGo procures directly

Contentsquare

Annual subscription (sessions)

GA4 360

Enterprise tier — required at our scale

WebEngage

Annual subscription (MAUs)

Adobe Target + OneTrust

Annual subscriptions

BigQuery + GCP

Pay-per-use (storage + queries)

CDP

Cloud (recommended — see rationale)

09

How CSQ+Axeno adds value beyond the brief

Innovative applications of the same platform investment — ideas IndiGo's evaluation committee may not have scoped, but that fall naturally out of the architecture already proposed.

Eight ideas the RFP didn't ask for — but the platform already supports

None of these require new licences beyond what's already proposed. Each reuses the same Contentsquare + GA4 + WebEngage + BigQuery ML foundation for a purpose beyond the 14 core capabilities.

01

Predictive disruption readiness

Blend historical IROPS patterns with weather/ATC signals in BigQuery ML to flag at-risk routes hours before a delay is declared — pre-staging WebEngage audiences so proactive outreach goes out before the reactive scramble starts.

02

Real-time next-best ancillary offer

Combine CS behavioural signals (what a passenger hovers over, not just what they click) with WE engagement history to serve the single highest-propensity ancillary — seat, bag, meal, lounge — at the moment of peak intent, not a generic upsell block.

03

Digital Experience Health Score

A composite KPI blending CS frustration signals + GA4 conversion + WE engagement into one number per route/segment/device. Requires a validated weighting methodology (a scoped data-science exercise, not a simple blend) — estimated as a 3-4 week workstream once approved.

04

Contact-centre agent-assist

Surface a customer's CS session replay and WE profile in the agent's screen when a call connects. Depends on IndiGo naming its CCaaS/telephony platform (Genesys, NICE, etc.) — no integration point exists for this today; feasibility to be confirmed once IndiGo's contact-centre stack is identified.

05

GenAI executive briefings

A weekly 'State of Digital' briefing in plain English, spanning CS + GA4 + WE data. This is a new cross-platform LLM orchestration layer reading from BigQuery — broader than 'extending Sense AI' (which only sees CS data) — scoped as its own small build.

06

Fraud & anomaly detection reuse

The same Sense AI anomaly detection and BigQuery ML infrastructure built for churn/CLV can be pointed at booking and payment patterns to flag fraud-adjacent anomalies — a second use case from infrastructure already being paid for.

07

Sustainability-conscious segment

Build a dedicated audience for environmentally-conscious travellers — surfacing direct-flight options and carbon-offset add-ons preferentially — a brand-differentiating segment increasingly relevant to corporate travel policy and ESG-driven B2B accounts.

08

Analytics Centre of Excellence

Beyond training sessions, establish a certification pathway so IndiGo's own analysts become internally self-sufficient over 12–18 months — reducing long-run dependency on any external partner, Axeno included.

Spotlight — the two nearest-term ideas (Month 7–9)

01 · Digital Experience Health Score — one number, three platforms, no manual stitching

Leadership currently checks three separate dashboards to gauge digital health: Contentsquare for frustration, GA4 for conversion, WebEngage for engagement. This blends the three into one composite score per route/segment/device. Honest scoping note: the blend needs a validated weighting methodology, not just an average — a dedicated 3–4 week data-science exercise before this becomes a trusted leadership metric, not a same-week toggle.

Built from: BigQuery as the unifying layer (already proposed) + a new weighting model validated jointly with IndiGo's analytics team — no new platform licence, but real analyst time required before launch.

02 · GenAI executive briefings — plain-English weekly digest, not a new dashboard to learn

A weekly 'State of Digital' summary in plain English — what moved, why, and what to do about it — delivered to leadership inboxes. Honest scoping note: this reads CS + GA4 + WE data together, so it's a new small LLM-orchestration build reading from BigQuery, not simply 'turning on' Contentsquare's Sense AI (which only sees Contentsquare's own data).

Built from: BigQuery (already proposed) feeding a lightweight LLM summarisation layer — scoped and estimated as its own small build once approved, not bundled free into the base platform cost.

When these ideas could be introduced

None of this displaces the core 26-week build. These are Year 1–2 conversations, sequenced after the foundation is live and stable.



How these get prioritised: each idea is scoped, estimated, and brought to IndiGo's Quarterly Business Review (Section 8 governance cadence) as a proposal with expected impact and cost — nothing here is assumed into scope or commercial value without IndiGo's explicit sign-off. This is a menu, not a commitment.

A

Technical appendix

Full-depth reference material for technical evaluators — Decision Framework justification, Integration Patterns detail, and complete Non-Functional Requirements mapping.

Not required reading for the business narrative; every claim traces back here.

Non-functional requirements — full detail

Requirement area	Our approach
Performance & scalability	CS processes 15B+ sessions/day globally. GA4 processes billions of events. All SaaS platforms auto-scale horizontally. Dashboard latency <3s. Billions of annual events supported.
Architecture & solution design	Cloud-native, microservices-based, API-first. Multi-environment deployment (Dev/QA/UAT/Pre-Prod/DR/Prod). HLA and LLA documents co-created with IndiGo during discovery.
Security, privacy & compliance	ISO 27001, 27017, 27018, 27701, SOC 2 Type II. AES-256 at rest, TLS 1.2+ in transit. RBAC, MFA, SSO (SAML/OAuth/OIDC). PII masking in session replay. Formal DPDP compliance declaration from Contentsquare.
Data residency	BigQuery on GCP Mumbai (asia-south1) serves as the India-resident system of record for all analytics data, unifying Contentsquare, GA4, and WebEngage exports. Combined with Contentsquare's contractual DPDP compliance framework and WebEngage's native India-hosting option.
Integration & data connectivity	120+ native integrations. REST APIs, Webhooks, Kafka/Kinesis streaming. CRM (Salesforce), CDP, BI (Tableau/Power BI), DWH (BigQuery/Snowflake). SDKs: Web, iOS, Android, React Native, Flutter.
SDK & implementation	Lightweight SDKs (<1% perf impact, per Contentsquare's published benchmark). Async event collection, offline queuing, remote config, kill switch. Backward compatibility. Adobe-to-GA4+CS migration led by Axeno.
Data management & retention	Configurable retention policies. Session replay: 13 months. GA4: 14 months (expandable via BQ). BigQuery: unlimited, with India data residency as described above.
Support & managed services	CS: Dedicated TAM + CSM, 24x7 Sev-1 support. Axeno: Noida-based team, Adobe migration, quarterly optimisation workshops, training in Hindi/English.

Why each architectural building block was selected

Building block	Extends / replaces	Why this choice — justification	Data flow
Contentsquare	Adobe Analytics (experience layer)	The only platform combining session replay, heatmaps, frustration detection, product analytics (autocapture), and AI-powered journey analysis in one stack. Used by Ryanair, easyJet, Thalys. Delivers behavioural context at a depth no adjacent tool matches — session replay shows why a metric moved, not just that it moved.	Client-side SDK → auto-captures all interactions → journey/funnel/DXA processing → Data Connect exports to BigQuery
Google Analytics 4	Adobe Analytics (marketing layer)	Industry standard for marketing attribution with 6 native models including data-driven. Native BigQuery export eliminates ETL. Predictive metrics (churn, purchase probability) are built in. Google Ads integration provides closed-loop campaign measurement.	gtag.js / Firebase SDK → real-time event stream → attribution processing → BigQuery Linking (native, zero-ETL)
WebEngage	(new capability)	Extends the stack with CDP-grade identity resolution and full-channel audience activation — push, email, SMS, WhatsApp, in-app — that completes the marketing and engagement picture. India-region hosting available. Predictive segmentation for engagement-driven forecasting.	WE SDK → event ingestion → profile unification → audience qualification → campaign activation → Data Export API to BigQuery
CDP (CDP / Middleware)	Adobe Launch / DTM	The event-middleware layer — a single, horizontally auto-scaling collection point with fan-out routing to all platforms, eliminating the $O(n^2)$ point-to-point integration problem. Tracking Plans enforce schema validation before events reach any platform; Identity Resolution provides cross-platform canonical user_id. See the dedicated slide on why this is kept alongside WebEngage's CDP, not replaced by it.	Server-side API + SDK events → schema validation → identity stitching → transformation → parallel routing to CS + GA4 + WE + persistence layer
Cloud Run (Microservices)	(new capability)	the CDP/event-pipeline layer's transformation logic covers event-shaping logic, but doesn't publish standalone APIs, serve real-time ML predictions, or run stateful multi-step orchestration. Cloud Run provides a small set of serverless Node.js functions for exactly those four jobs — scaling to zero between calls, consistent with the serverless-first design used everywhere else in this architecture.	3Scale Gateway routes external API calls → Cloud Run function executes business logic → reads/writes Cloud SQL or calls BigQuery ML for real-time scores

Why each architectural building block was selected

Building block	Extends / replaces	Why this choice — justification	Data flow
BigQuery (GCP Mumbai)	Adobe Data Warehouse	Delivers the India data residency requirement (Req 7.2.1) natively via the asia-south1 region. Serverless, pay-per-query economics. Native GA4 integration (BigQuery Linking). Unified analytical layer enabling cross-platform analytics beyond any single platform's native scope. BigQuery ML provides custom model training without separate infrastructure.	Receives exports from CS (Data Connect) + GA4 (BQ Linking) + WE (API) + CDP (direct) → unified star schema → Looker/Power BI queries → Reverse ETL to WebEngage
Cloud SQL + Memystore	(new capability)	BigQuery isn't built for single-record lookups — Cloud SQL provides fast transactional persistence for the middleware's operational state (identity mapping, config), while Memystore caches dedup/rate-limit checks that run on every event. Together they complete a three-tier model matched to real access patterns.	CDP checks Memystore (dedup/rate-limit) → writes operational state to Cloud SQL → streams analytical events to BigQuery — all three in GCP Mumbai
GTM + OneTrust + Adobe Target + DataHub	Adobe DTM, manual consent, manual catalogue	GTM: free, proven tag governance with server-side tagging, version control, maker-checker workspaces. OneTrust: consent management for DPDP (India, primary requirement) extending to GDPR for EU-origin passenger data on IndiGo's European routes — consent signals gate all data collection. Adobe Target: experimentation platform with native CS+GA4 integration, strong India presence. DataHub: open-source data lineage completing the metadata/glossary layer.	GTM manages SDK deployment lifecycle → OneTrust feeds consent signals to all SDKs → Adobe Target's experiments integrate with CS analysis → DataHub tracks lineage across the full pipeline
Cloud Composer + Secret Manager	(new capability)	Every batch job here — CRM sync, tier recalculation, ML retraining, DataHub crawls, the Adobe migration — needs one scheduler with dependency management and failure alerting, not scattered cron jobs. Cloud Composer (managed Airflow) is that single pane of glass. Secret Manager closes the parallel gap: every credential gets one governed, auto-rotating vault.	Cloud Composer triggers/monitors every scheduled batch job → Secret Manager supplies credentials to CDP, Cloud SQL, Cloud Run, and every GCP service, with rotation logged centrally

ARCHITECTURE PRINCIPLES

Separation of concerns — each platform does what it does best · API-first — every component replaceable without rearchitecting · Source-system abstraction — CDP decouples sources from destinations · India data residency — BigQuery Mumbai ensures DPDP alignment regardless of platform processing location · Event-driven — async, non-blocking, horizontally scalable · Serverless-first — no component requires standing infrastructure to patch or capacity-plan

Integration patterns — real-time vs. batch data exchange

Not every integration needs sub-second latency, and forcing everything real-time adds cost and fragility without business benefit. Each integration point below is deliberately classified by the latency its use case actually requires.

Integration point	Mode	Latency target	Mechanism
Website/App SDK events (CS, GA4, WE)	Real-time	< 1 second	Client SDK → CDP middleware → parallel fan-out to all destinations
Payment gateway webhook	Real-time	< 2 seconds	Server-side webhook → CDP server-side API
Booking engine / PSS events	Real-time	< 2 seconds	Server-side API → CDP → BigQuery + platforms
IROPS / operational (OPS) feed	Real-time	< 5 seconds	CDP server-side webhook → BigQuery → WebEngage audience trigger
Loyalty points accrual	Real-time	< 5 seconds	CDP server-side API → WebEngage profile update
WebEngage journey triggers (push/email/SMS/WA)	Real-time	Seconds	WebEngage journey orchestrator, event-triggered
Chatbot conversation events	Real-time	< 2 seconds	Webhook → CDP server-side API
Real-time offer/scoring API calls	Real-time	< 500ms	Cloud Run microservice → BigQuery ML / Cloud SQL lookup, synchronous response
GA4 → BigQuery (streaming export)	Near real-time	Minutes	GA4 BigQuery Linking, streaming mode
Contentsquare → BigQuery (Data Connect)	Batch	Daily	Scheduled export job
WebEngage → BigQuery (Data Export API)	Batch	Daily (2–4x/day)	Scheduled API pull
Salesforce CRM sync	Batch	Nightly	Cloud Composer-orchestrated incremental sync via CDP
BigQuery ML model retraining (churn, CLV)	Batch	Weekly	Cloud Composer-scheduled BigQuery ML pipeline
Loyalty tier recalculation	Batch	Nightly	Cloud Composer-scheduled aggregation job over accrual events
Adobe historical data migration	Batch (one-time)	12-month backload, Phase 4	Adobe Data Feed export → BigQuery bulk load
DataHub metadata/lineage sync	Batch	Daily	Cloud Composer-scheduled catalogue crawl
Looker / Power BI report refresh	Batch	Per report cadence (hourly–daily)	Scheduled dashboard refresh

Integration mode mapped to the eight in-scope journeys

Cross-referencing Section 3.1's user journeys against the integration patterns on the previous page — most journeys blend both modes deliberately.

#	Journey (Section 3.1)	Real-time integration used	Batch integration used
1	Booking funnel (Search → Confirmation)	SDK events, payment webhook, booking engine events — every step is real-time	Post-booking GA4/BigQuery aggregation for reporting
2	Abandoned booking recovery	Drop-off detection, root-cause classification, re-engagement trigger	Audience segment refresh (near real-time, minutes)
3	Web & app check-in	Check-in events, boarding pass generation, travel-day push	T-48h reminder trigger (scheduled batch job over booking data)
4	IROPS & flight disruption management	OPS feed ingestion, affected-passenger notification — time-critical, real-time only	Post-disruption impact analysis (batch, next-day)
5	Marketing campaign lifecycle	Campaign click/landing attribution capture	Campaign ROI rollup, CLV modelling (batch, daily/weekly)
6	BluChip loyalty programme lifecycle	Points accrual, milestone notifications	Tier recalculation, churn-risk scoring — BigQuery ML (batch, nightly)
7	Flash sale & peak traffic management	Live funnel monitoring, staggered notification, anomaly alerts	Post-sale comprehensive analysis (batch, next-day)
8	Experimentation & CRO	Adobe Target variant serving to live traffic (real-time)	Heatmap/session analysis, revenue-impact calculation (batch)

Design principle: real-time is used only where the business moment is time-sensitive (a payment, a disruption, a live monitoring window) — everything else runs batch, which is cheaper to operate and easier to reconcile.

In Summary

CSQ as the core platform with GA4 & Webengage. Fourteen capabilities, fully covered.

Contentsquare's behavioural depth, GA4's attribution standard, and WebEngage's engagement reach - delivered by Axeno with complete accountability and a governed 26-week delivery timeline.

14/14 capabilities

26-week delivery

Best-in-class capabilities

1 accountable partner

Commercial details will be provided once sizing information is shared